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BR-DETR: a lightweight real-time detector for multi-scale concrete bridge defect inspection from UAV imagery

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ABSTRACT

Accurate quantification of concrete bridge defects, including category classification, spatial localization, and geometric measurement, is fundamental to structural health monitoring, since condition assessment relies directly on the localization and dimensional accuracy of the detection system. Unmanned aerial vehicle (UAV) inspection offers a promising acquisition modality, but deploying deep-learning detectors on edge devices faces three challenges: defect morphological diversity, limited resolution of small-scale damage, and constrained onboard computation. This paper proposes BR-DETR, a lightweight variant of RT-DETR-R18 with three task-specific components. The C2f-DIMB backbone employs dynamic multi-branch depthwise convolutions with heterogeneous kernel geometries to adapt the receptive field to diverse defect morphologies while reducing backbone parameters. The Multi-Scale Small Defect Enhanced Pyramid (MSDEP) injects fine-grained P2-level information into the feature pyramid to improve small-defect sensitivity without adding a detection head. The AIFI-HiLo encoder decomposes self-attention into frequency-specialized branches for high-frequency boundaries and low-frequency structural context. On a five-class dataset of 8,924 images, BR-DETR attains 72.51% mAP50 and 54.36% mAP50-95, improving over RT-DETR-R18 by 4.67 and 5.67 percentage points while reducing parameters by 31.9% to 13.54 M and GFLOPs by 8.6%. Considered as a measurement instrument, it reduces the mean center localization error from 6.83 px to 4.92 px (−28.0%) and raises the share of high-quality detections (IoU $\geq$ 0.75) from 52.3% to 61.8%. Deployed on an NVIDIA Jetson Xavier NX, it achieves 46 FPS with 21.7 $\pm$ 1.2 ms latency and a memory footprint below 700 MB, providing a practical tool for real-time, quantitative UAV-based bridge condition assessment.

1. Introduction

Concrete bridges represent the majority of global highway and railway infrastructure due to their durability and cost-effectiveness. However, under prolonged exposure to moisture, temperature fluctuations, and chemical corrosion, they inevitably suffer from structural defects [1], including cracks, spallation, efflorescence, exposed reinforcement bars, and corrosion, as illustrated in Fig. 1. Without timely detection and repair, these defects can lead to catastrophic failures that endanger public safety and incur substantial economic losses. Traditional inspection relies on manual visual examination, which is time-consuming, sometimes requires traffic disruption or special access equipment [2], and is inherently subjective, leading to inconsistent severity evaluation and missed detections [3]. The growing backlog of aging bridges therefore demands automated, efficient, and reliable inspection technologies.

From a measurement standpoint, accurate detection and localization constitute the fundamental measurement task of bridge inspection. In particular, mAP50-95 serves as a proxy for geometric measurement precision, since high precision at strict thresholds (IoU $\geq$ 0.75) bounds the deviation of predicted box boundaries to a small fraction of the object. Box dimensions translated through the GSD further yield physical-size estimates such as crack length and spallation area, so accurate quantification of concrete bridge defects, including their category, location, and geometric extent, is fundamental to structural health monitoring.

Deep learning has provided the preconditions for automated structural health monitoring [4], and object detection frameworks are particularly attractive because they simultaneously localize and classify multiple defects in a single image. Two-stage detectors such as Faster R-CNN [5] achieve high accuracy, but their computational cost limits real-time deployment. One-stage detectors such as the YOLO series [6–9]

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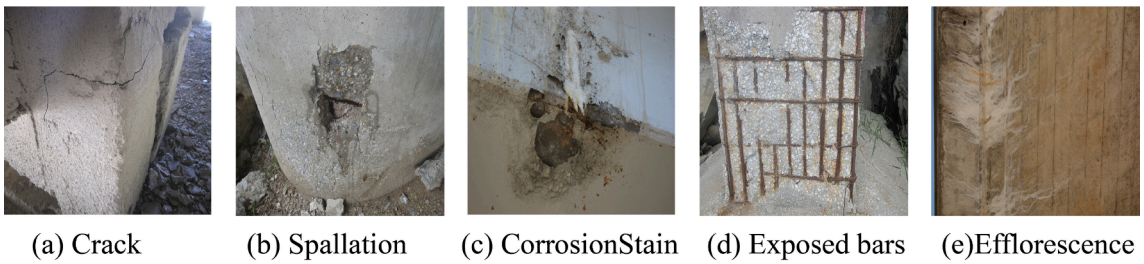


Fig. 1. Five types of concrete bridge damage.

enable real-time inference but rely on anchor boxes and non-maximum suppression (NMS), which impede end-to-end deployment. Recently, RT-DETR [10] represents the first real-time end-to-end Transformer detector, combining the accuracy of DETR-based methods with YOLO-level inference speed through an efficient hybrid encoder. UAVs further offer a superior acquisition platform that avoids the risks of high-altitude human work and flexibly reaches inaccessible areas. Modern flight-control strategies, such as fixed-time prescribed-performance trajectory tracking [11], reliability-constrained attitude tracking [12], and fixed-time attitude control under stochastic disturbances [13], enable quadrotor UAVs to maintain stable imaging postures even under wind disturbance. The primary challenge of UAV-based inspection has thus shifted from platform stability to the onboard visual perception system, which is the focus of this work.

UAV imaging nevertheless introduces challenges that translate directly into detection difficulties. First, the ground sampling distance (GSD) varies with the camera-to-surface distance, so identical physical defects appear at pixel scales differing by a factor of up to five, demanding scale-adaptive feature extraction. Second, motion blur from platform vibration attenuates exactly the high-frequency edge gradients on which crack boundary delineation relies. Third, uneven illumination and self-shadowing further suppress the already-low contrast of defects such as efflorescence, so reliable recognition must draw on low-frequency global context. Fourth, oblique viewing angles distort nominally horizontal or vertical linear structures into arbitrarily oriented ones, requiring receptive fields that can compose arbitrary orientations rather than relying on axis-aligned kernels. These four imaging-level difficulties motivate the three architectural responses of this paper, namely dynamic receptive-field adaptation, fine-grained resolution enhancement, and frequency-decoupled attention, establishing a clear problem-to-solution correspondence for each architectural component. Moreover, the computational demands of conventional detectors make them difficult to deploy on UAV edge hardware.

This study adopts RT-DETR-R18 as the baseline owing to its NMS-free end-to-end architecture, compact size, and real-time capability on embedded platforms. Nevertheless, its direct application to bridge defects reveals three limitations: (i) the fixed-kernel ResNet-18 backbone cannot adaptively accommodate the morphological diversity of defects, from elongated cracks to compact spallation; (ii) the standard three-scale feature pyramid progressively attenuates fine-grained spatial information, degrading sensitivity to small-scale cracks and corrosion; and (iii) the vanilla self-attention in the encoder insufficiently distinguishes high-frequency edge details from low-frequency structural context. To overcome these limitations, the main contributions are summarized as follows:

(1) To address the geometric mismatch between fixed-kernel backbones and the heterogeneous morphology of bridge defects, a lightweight C2f-Dynamic Inception Mixer Block (C2f-DIMB) backbone is designed to replace the parameter-heavy ResNet-18 with dynamic multi-branch depthwise convolutions. Its core novelty is the Dynamic Kernel Weighting (DKW) mechanism, which adaptively re-weights heterogeneous kernel geometries (square, horizontal-band, and vertical-band)

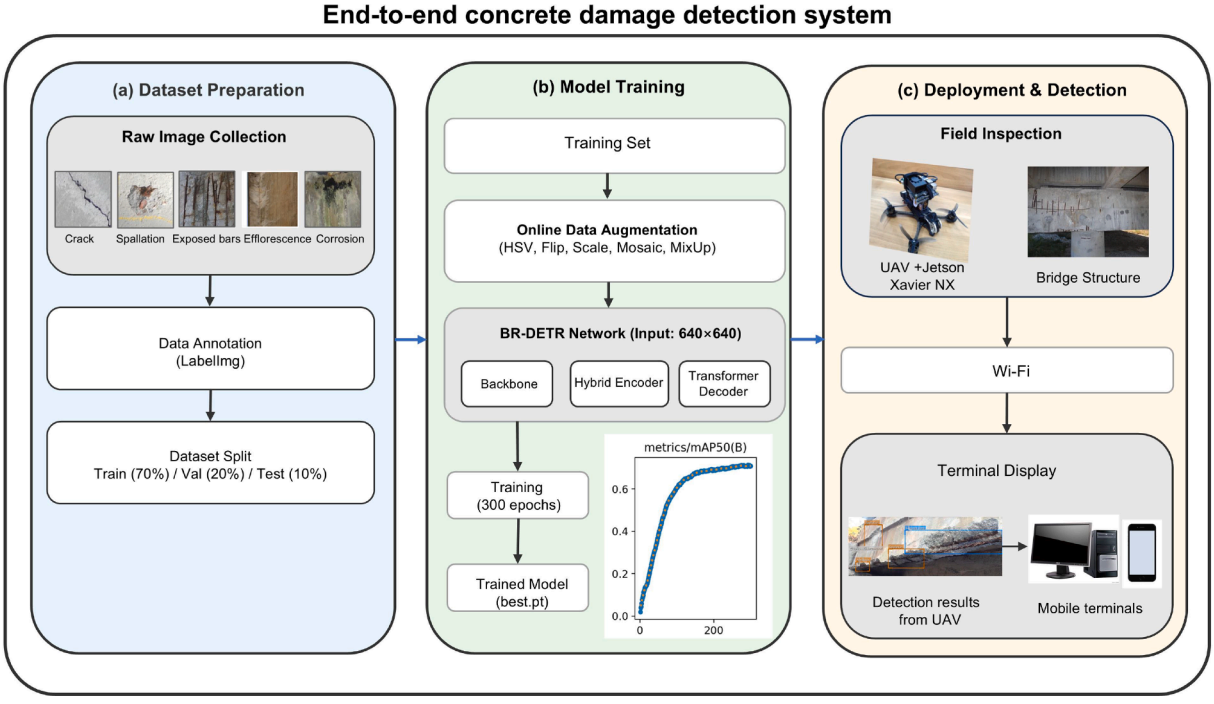


Fig. 2. End-to-end Concrete Damage Detection System.

according to input content — an input-conditional receptive-field adaptation that, to the best of our knowledge, has not previously been exploited for structural defect detection.

(2) A Multi-Scale Small Defect Enhanced Pyramid (MSDEP) is designed to counter the progressive spatial attenuation of small defects. MSDEP injects fine-grained P_2-level information into the feature pyramid through a computationally efficient pathway, improving small-defect sensitivity and boundary precision without an additional detection head. Compared with directly attaching a P_2 head, MSDEP attains a comparable small-defect gain at approximately one-quarter of the additional FLOPs while preserving embedded real-time throughput. This directly reduces localization error for small defects, which is critical for damage-area measurement.

(3) An AIFI-HiLo encoder is developed, in which self-attention is decomposed into frequency-specialized branches that separately process high-frequency boundary information and low-frequency structural context. AIFI-HiLo preserves the edge gradients essential for crack quantification while capturing the contextual cues required for ambiguous defects such as efflorescence, at roughly 31% of the computational cost of standard MHSA. Unlike a direct transplantation of HiLo attention, the head-split ratio and window configuration are selected from the spectral statistics of bridge defects measured by 2D-FFT analysis, grounding the decomposition in the imaging characteristics of the task.

As illustrated in Fig. 2, the resulting end-to-end inspection system comprises three stages: dataset preparation (multi-source images cleaned, re-annotated under a unified five-category taxonomy, and split 7:2:1), model training (300 epochs with online augmentation, optimal weights exported to a TensorRT FP16 engine), and onboard deployment (the engine runs on a Jetson Xavier NX mounted on the UAV and processes camera frames in real time, relaying category, position, and confidence to ground terminals over Wi-Fi). BR-DETR improves mAP50 by 4.67 percentage points (pp) and reduces mean center localization error by 28.0% over the baseline, while maintaining real-time embedded inference at 46 FPS on the Xavier NX.

The remainder of this paper is organized as follows. Section 2 reviews related work; Section 3 details the proposed BR-DETR; Section 4 describes the experimental setup; Section 5 presents the results and analysis; Section 6 concludes the paper.

2. Related work

This section reviews object detection for bridge defect detection, attention mechanisms, and multi-scale feature fusion. Beyond visual detection, deep learning now underpins a broad range of data-driven industrial condition-monitoring tasks, such as interpretable remaining-useful-life prediction [14] and source-free domain-adaptive fault diagnosis of power converters [15]. The present work focuses on the visual detection branch for concrete bridge structures.

Object detection for bridge defects has progressed through three paradigms. **Two-stage detectors** follow a region-proposal-and-classification pipeline, most representative Faster R-CNN [5]; in bridge applications, Xu et al. [16] applied a region-based deep convolutional network for concrete damage identification, and Li et al. [17] deployed Faster R-CNN on UAV platforms for crack detection; however, the sequential proposal–classification pipeline inherently limits inference speed for real-time UAV deployment.

One-stage detectors regress bounding boxes directly from dense feature maps. SSD [18] introduces multi-scale anchor-based prediction and has been applied to bridge surface defect detection [19], while RetinaNet addresses class imbalance through focal loss. The YOLO series advances real-time performance through successive architectural optimizations [6–9], with the most recent versions, YOLOv12 [20] and YOLOv13 [21], adopting attention-centric and hypergraph-enhanced designs. For bridge defects, Zhang et al. [22] proposed an attention-enhanced single-stage model, Jiang et al. [23] developed a lightweight multi-scale network for multi-damage recognition in complex environments, and Chen and Jiang [24] proposed CD-YOLOv8 for real-time embedded bridge inspection. Nevertheless, anchor-based one-stage detectors still depend on NMS post-processing, which impedes end-to-end deployment.

Transformer-based detectors represent a paradigm shift. DETR [25] formulates detection as set prediction, eliminating anchors and NMS; Deformable DETR [26] accelerates convergence and improves small-object detection, DAB-DETR [27] reformulates queries as dynamic anchor boxes, and DINO-DETR [28] reaches state-of-the-art accuracy via contrastive denoising. More recently, RT-DETRv2 [29], D-FINE [30], and DEIM [31] further push the accuracy–latency frontier of end-to-end detection through training refinements, fine-grained box-distribution regression, and improved matching, respectively; however, these general-purpose detectors are not tailored to the scale statistics, low-contrast categories, and embedded constraints of UAV bridge-defect imagery. Among RT-DETR adaptations for bridges, Yu et al. [32] proposed RTDetrCrack for cloud–edge crack detection and Dai et al. [33] inserted CBAM attention into the RT-DETR encoder, but both are limited to single-class crack detection. BR-DETR addresses morphological diversity, small-defect sensitivity, and frequency-aware encoding within a unified multi-class framework.

Attention mechanisms emphasize discriminative features while suppressing background interference — valuable on concrete surfaces exhibiting shadows, stains, and construction joints. Channel and spatial attention designs such as SENet [34] and CBAM [35] have been widely adopted for structural defect detection, but they do not differentiate spatial frequency components, limiting their capacity to handle high-frequency crack boundaries and low-frequency large-area defects simultaneously. Standard multi-head self-attention (MHSA) likewise allocates computation uniformly across all frequencies, which is problematic because high-frequency components carry the edge gradients critical for crack delineation while low-frequency components encode the context needed to classify low-contrast defects. LITv2 [36] decomposes attention into high- and low-frequency branches, offering a principled basis for frequency-aware modeling; however, it is a classification backbone whose fixed decomposition ratio is not optimized for multi-class bridge defects. AIFI-HiLo is therefore proposed in this work, integrating frequency-decoupled attention with a task-adapted head-split configuration directly into the RT-DETR encoder.

Multi-scale feature fusion is essential because defects range from hairline cracks of a few pixels to large-area spallation, and small-defect boundary precision directly determines the reliability of inspection measurements. FPN [37] establishes top-down fusion with lateral connections, PANet [38] adds bottom-up augmentation, BiFPN [39] introduces learnable fusion weights, and recent work improves cross-scale propagation through energy-preserving normalization and channel attention [40]. Small defects nonetheless remain challenging for standard P3–P5 pyramids; adding a high-resolution P2 detection head recovers them but significantly increases computation and post-processing time, which is impractical on resource-constrained devices. The rethinking-FPN work [41] introduces SNI and GSConvE for better cross-layer propagation, the MFM module [42] resolves cross-scale feature conflicts through dynamic modulation, the OmniKernel module [43] combines global, large-kernel, and local branches, and HS-FPN [44] screens high-level semantics before fusion to strengthen small-object responses at low cost. Moreover, the morphological similarity between defect types such as cracks and exposed bars leads to category confusion that existing fusion methods do not adequately address. Beyond average accuracy, measurement-oriented inspection must additionally account for localization uncertainty and cross-domain bias; recent measurement-focused studies emphasize quantified localization error and reliability under variable operating conditions [14,15]. Because multi-source public datasets differ in annotation standards, resolution, and illumination, a single random split can overestimate deployable performance — a perspective this work adopts through explicit measurement metrics, leave-one-source-out validation, and simulated imaging perturbations.

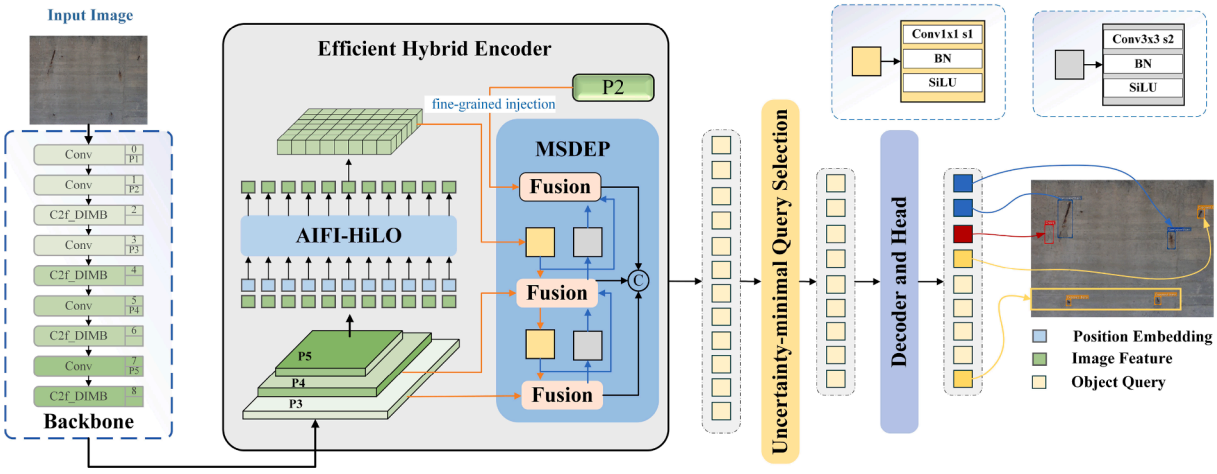


Fig. 3. Overall architecture of the proposed BR-DETR framework.

In summary, three limitations motivate BR-DETR: (a) fixed-kernel backbones cannot adaptively capture diverse defect morphologies; (b) existing attention mechanisms treat all frequency components uniformly; and (c) current pyramids either sacrifice efficiency with P2 heads or miss small defects without them.

3. Methodology

This section first introduces the baseline RT-DETR-R18 model and the rationale for its selection, then explicitly states the origin and novelty of each component, describes the overall architecture of our proposed BR-DETR, and finally details the three new modules.

3.1. Overall architecture of BR-DETR

RT-DETR is the first real-time end-to-end Transformer detector, formulating detection as set prediction and thus removing the latency and threshold sensitivity of NMS post-processing. The RT-DETR-R18 variant employs a ResNet-18 [45] backbone producing multi-scale features $\{P_3, P_4, P_5\}$ at strides of 8, 16, and 32, refined by a hybrid encoder consisting of an Attention-based Intra-scale Feature Interaction (AIFI) module and a Cross-scale Feature Fusion (CCFF) module, and decoded by a Transformer decoder with learnable object queries. Its direct application to UAV bridge-defect imagery, however, exhibits the three limitations identified above: fixed 3×3 kernels lacking geometric adaptability to defect morphology, insufficient enhancement of small defects whose spatial information is diluted by downsampling, and frequency-agnostic self-attention in the AIFI encoder. As illustrated in Fig. 3, BR-DETR addresses these limitations through three targeted improvements while preserving the NMS-free end-to-end paradigm:

(1) **C2f-DIMB backbone**: replaces ResNet-18 with dynamic multi-branch depthwise convolutions that adaptively adjust receptive field geometry according to defect morphology;

(2) **Multi-Scale Small Defect Enhanced Pyramid (MSDEP)**: injects fine-grained P2-level spatial information into the feature pyramid through a computationally efficient pathway, improving small defect detection and boundary delineation without adding an extra detection scale;

(3) **AIFI-HiLo encoder**: decomposes self-attention into frequency-specialized high and low branches to separately preserve defect boundary sharpness and global structural context for low-contrast defect classification.

The three modules are complementary rather than additive, and each builds a task-specific formulation on top of well-established primitives. C2f-DIMB adapts receptive-field geometry in the spatial domain through the new DKW weighting, MSDEP enhances resolution fidelity in the

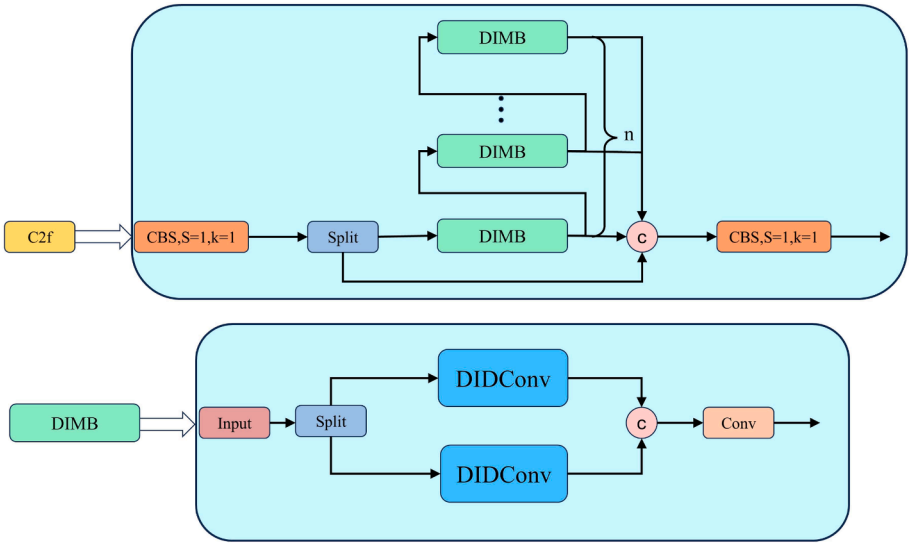


Fig. 4. The C2f-DIMB module.

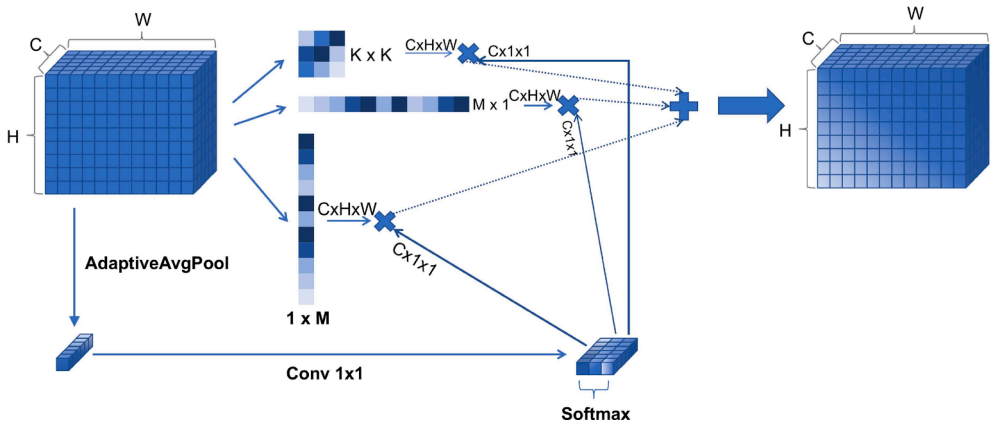


Fig. 5. Structure diagram of the DIDConv module.

pyramid through an architecture-level composition that injects P_2 information without an extra head, and AIFI-HiLo decouples frequency content in the encoder through a spectrum-driven head split. Each targets a failure mode the others cannot address, as the pairwise ablation study confirms.

3.2. C2f-DIMB backbone

The fixed 3 × 3 kernels of ResNet-18 present a fundamental mismatch with bridge-defect morphology: crack lengths may exceed 500 pixels with extreme aspect ratios that square kernels capture inefficiently, whereas spallation and corrosion regions of 20 × 20 to 400 × 400 pixels with irregular boundaries require isotropic receptive fields. Therefore, the Dynamic Inception Mixer Block (DIMB) is designed and embedded into the C2f framework of YOLOv8 [46] to construct the C2f-DIMB backbone, which comprises four stages producing $\{P_2, P_3, P_4, P_5\}$ at strides of 4, 8, 16, and 32; its detailed structure is depicted in Fig. 4.

The C2f module provides an efficient split-and-concatenate architecture for gradient-rich feature learning. Given an input $\boldsymbol{X} \in \mathbb{R}^{C\times H\times W}$, the module first projects and splits the features along the channel axis, then passes one branch through a sequence of n DIMB blocks while preserving the other branch as an identity shortcut:

$$\boldsymbol{X}_0,\ \boldsymbol{X}'_0 = \mathrm{Split}(\mathrm{Conv}_{1\times1}(\boldsymbol{X})),\quad \boldsymbol{X}_i = F^{(i)}_{DIMB}(\boldsymbol{X}_{i-1}),\ i = 1,\ldots,n \tag{1}$$

$$\boldsymbol{Y} = \mathrm{Conv}_{1\times1}(\mathrm{Concat}(\boldsymbol{X}'_0,\ \boldsymbol{X}_0,\ \boldsymbol{X}_1,\ \ldots,\ \boldsymbol{X}_n)) \tag{2}$$

where the prime symbol (′) denotes the identity branch retained without DIMB processing. This design ensures that both unprocessed shallow features and progressively refined deep features contribute to the output, facilitating stable gradient propagation.

To capture structural damage features with diverse orientations and scales, a Dynamic Inception Depthwise Convolution (DIDConv) module is designed, as illustrated in Fig. 5. The core computational unit of DIMB, designed to provide input-adaptive multi-scale spatial feature extraction. For an input $\boldsymbol{Z} \in \mathbb{R}^{C\times H\times W}$, three parallel depthwise convolution branches with distinct kernel shapes are applied:

$$\boldsymbol{Z}_j = \mathrm{DWConv}_{k_j}(\boldsymbol{Z}),\quad j \in \{sq, h, v\} \tag{3}$$

where the kernel configurations are $k_{sq} = k \times k$(square), $k_h = 1 \times M$(horizontal band), and $k_v = M \times 1$ (vertical band), with $M = 3k+2$ and default $k = 3$ The square branch captures isotropic features suited for compact spallation regions and corrosion patches, while the horizontal and vertical band branches provide elongated receptive fields aligned with cracks and exposed rebar that may orient along gravity or structural member axes. In addition to the three directional branches, a global context branch is incorporated via AdaptiveAvgPool followed by a 1 × 1 convolution, which extracts channel-wise global statistics to complement local structural features. The outputs of all branches are subsequently projected by individual C × 1 × 1 convolutions to produce attention weights, which are normalized via Softmax and used to perform a weighted fusion of the branch features, yielding the final output feature map of the same spatial dimension C × H × W.

To enable input-adaptive branch selection, we introduce a Dynamic Kernel Weighting (DKW) mechanism that generates content-dependent attention weights through global channel context:

$$\boldsymbol{\omega} = \mathrm{Softmax}(\mathscr{R}(\mathrm{Conv}_{1\times1}(\mathrm{GAP}(\boldsymbol{Z})))) \in \mathbb{R}^{3\times C\times1\times1} \tag{4}$$

$$\widehat{Z} = \sigma\left(BN\left(\sum_{j\in\{sq,h,v\}} \omega_j \odot Z_j\right)\right) \tag{5}$$

where GAP(·) denotes global average pooling, $\mathscr{R}(\cdot)$ denotes the reshape operation that rearranges the 3C-dimensional vector into a $3 \times C$ weight matrix, and $\sigma(\cdot)$ is the SiLU activation. Through this mechanism, the network learns to dynamically emphasize elongated receptive fields for crack-dominant regions while prioritizing the isotropic branch for area-type defects such as spallation.

Coverage of arbitrary orientations. It is worth clarifying that the horizontal (1 × M) and vertical (M × 1) band kernels are not intended to literally enumerate all crack orientations. Rather, together with the square branch they span a basis of anisotropic receptive fields whose convex combination — under the Softmax-normalized DKW weights — can approximate arbitrary orientations: for a diagonal crack, the network activates the horizontal and vertical branches simultaneously ($\omega_h \approx \omega_v > \omega_{sq}$), and the effective receptive field of the weighted sum elongates along the diagonal. Section 5.7 verifies this behavior empirically on 45° and irregularly oriented cracks.

To further broaden the receptive field diversity, the Dynamic Inception Mixer module splits input channels into two groups and processes each with a DIDConv at different kernel scales ($k = 3$ and $k = 5$), yielding six distinct effective receptive field patterns from a single module. This multi-scale coverage is particularly beneficial for corrosion detection, where the gradual color transition requires larger contextual windows, and for distinguishing exposed rebar from surface cracks based on width and texture differences.

Each C2f-DIMB block arranges the DIM token-mixer and a Convolutional Gated Linear Unit (ConvGLU) [47] feed-forward network in a dual-residual structure with learnable layer scaling:

$$\boldsymbol{X}'' = \boldsymbol{X} + \gamma_1 \cdot \mathrm{DIM}(\mathrm{BN}(\boldsymbol{X})) + \gamma_2 \cdot \mathrm{ConvGLU}(\mathrm{BN}(\boldsymbol{X}')) \tag{6}$$

where $\gamma_1, \gamma_2 \in \mathbb{R}^C$ are learnable scale initialized to 10^{-2}, stabilizing early training; the ConvGLU split-gate mechanism enhances channel interactions that help separate visually similar pairs such as crack versus construction joints. Compared with ResNet-18, C2f-DIMB thus offers

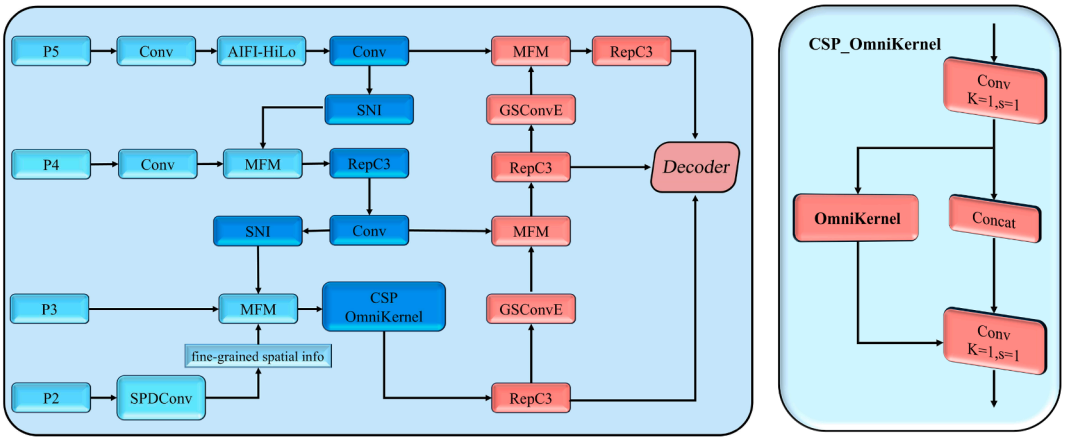


Fig. 6. The architecture of MSDEP.

morphology-adaptive receptive fields without manual kernel selection, a substantially reduced parameter count through extensive depthwise convolution, and complementary geometric priors for defects with diverse aspect ratios, from elongated cracks to compact spallation regions.

3.3. Multi-Scale small defect enhanced pyramid (MSDEP)

In the standard RT-DETR pyramid, a 16-pixel defect corresponds to only a 2 × 2 activation at the finest P_3 level, so small defects are disproportionately degraded by downsampling, and even medium defects lose boundary detail. Attaching an additional P_2 detection head recovers resolution but inevitably increases computation, conflicting with real-time UAV deployment. As illustrated in Fig. 6, MSDEP instead enhances representation within the existing three-scale framework, jointly addressing fine-grained information preservation, adaptive cross-scale interaction, and stable inter-scale transfer:

$$\{\widehat{P}_3, \widehat{P}_4, \widehat{P}_5\} = \mathscr{M}_{\mathrm{MSDEP}}(P_2, P_3, P_4, P_5) \tag{7}$$

First, to preserve fine-grained spatial information that is essential for small defect perception, a dedicated enhancement pathway is introduced to inject high-resolution details from shallow features into the feature fusion stage. Conventional strided convolution and pooling operations inevitably discard spatial details during resolution reduction, which disproportionately affects small defects. To mitigate this issue, SPDConv [48] is employed to transform spatial information into channel-wise representations via a space-to-depth operation:

$$\mathscr{S}_{\downarrow s} : \mathbb{R}^{C\times H\times W} \mapsto \mathbb{R}^{s^2C\times \frac{H}{s}\times \frac{W}{s}} \tag{8}$$

By applying this transformation with $s = 2$ followed by channel projection, yields the resolution-matched P_2 feature:

$$\boldsymbol{F}^{\downarrow}_{P_2} = \phi_{1\times 1}(\mathscr{S}_{\downarrow 2}(P_2)) \tag{9}$$

This design enables the retention of fine boundary and texture information without introducing additional detection scales. The enhanced feature is then fused with both the top-down semantic features and the backbone P_3 feature at the corresponding level yielding:

$$\widehat{P}_3 = \Psi\left(\boldsymbol{F}^{\downarrow}_{P_2},\ \boldsymbol{F}^{\uparrow}_{P_4},\ P_3\right) \tag{10}$$

A CSP-OmniKernel module refines the fused features: OmniKernel captures multi-frequency representations through Fourier-domain global, large-kernel regional, and local branches, while the CSP wrapper restricts the operator to a 1/4 channel slice, bounding its overhead at the high-resolution fused node. Second, at each fusion node the multi-scale feature mixer (MFM) [42] computes channel-wise attention to adaptively balance scale contributions, alleviating scale imbalance without degrading large-scale representation. Finally, because nearest-neighbor upsampling replicates each value s^2 times and artificially inflates feature magnitudes after summation — systematically suppressing low-magnitude small-defect responses — an energy-preserving normalization (SNI) [41] is applied:

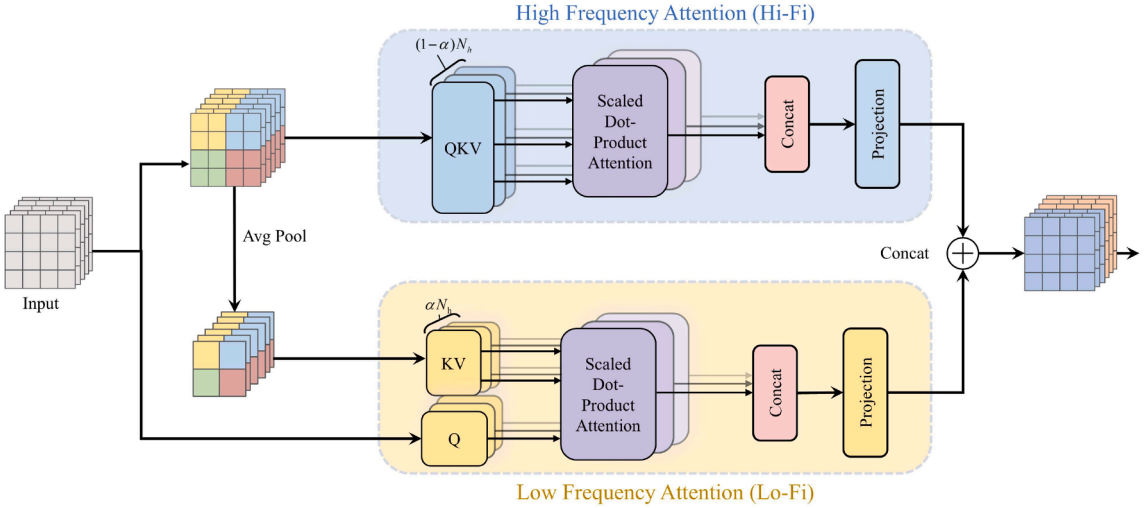


Fig. 7. The structure of AIFI-HiLo.N_h refers to the total number of self-attention heads at this layer. α denotes the split ratio for high/low frequency heads.

$$F^{\uparrow} = s^{-2}\cdot \mathrm{Upsample}_s(F) \tag{11}$$

MSDEP thus delivers an enhanced three-scale feature set to the decoder, improving small-defect sensitivity and boundary localization without an additional detection scale.

3.4. AIFI-HiLo: frequency-aware transformer encoder

The AIFI encoder applies standard MHSA to the P_5 map, allocating computation uniformly across spatial frequencies. This is doubly suboptimal for bridge defects: the high-frequency edge gradients critical for crack delineation are diluted by joint processing with low-frequency content, while the low-frequency context needed for low-contrast categories is obscured by surface-texture noise and UAV motion blur.

The frequency decomposition adopted here is grounded in the measured spectral statistics of the five defect categories rather than assumed a priori. 2D-FFT analysis of ground-truth defect patches (Fig. 16) shows that crack patches concentrate the majority of their spectral energy in the high-frequency band, reflecting their sharp, repetitive edge gradients, whereas spallation and efflorescence concentrate roughly two thirds of their energy in the low-frequency band, consistent with their diffuse, area-type appearance; corrosion and exposed bars exhibit intermediate spectra. This bimodal structure implies that any uniform attention allocation is suboptimal for at least one end of the spectrum, motivating the assignment of window-based local attention (a high-pass-like operator) to one head group and pooled global attention (a low-pass-like operator) to the other.

AIFI-HiLo accordingly partitions the attention heads into two frequency-specialized groups, as shown in Fig. 7.

Hi-Fi branch. A proportion $(1-\alpha)$ of attention heads apply window-based self-attention within non-overlapping local windows of size $w \times w$:

$$\text{Hi - Fi}(F) = \text{W - Attn}_w(Q_{hi}, K_{hi}, V_{hi}) \tag{12}$$

Within a small spatial window, low-frequency components vary slowly; therefore, local attention tends to emphasize high-frequency edge gradients that determine crack detection precision.

Lo-Fi branch. The remaining α proportion of heads perform global attention on spatially down sampled key and value representations:

$$\text{Lo - Fi}(F) = \text{Attn}(Q_{lo}, \text{Pool}_s(K_{lo}), \text{Pool}_s(V_{lo})) \tag{13}$$

The pooling operation acts as a low-pass filter that suppresses high-frequency noise from concrete surface texture and UAV-induced motion blur, while reducing attention complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N^2/s^2)$.The outputs of both branches are concatenated and linearly projected:

$$\text{HiLo}(F) = \phi_{1\times1}([\text{Hi - Fi}(F); \text{Lo - Fi}(F)]) \tag{14}$$

This frequency-level decoupling complements the spatial adaptation of C2f-DIMB and the resolution enhancement of MSDEP, completing the multi-level multi-scale design of BR-DETR.

3.5. Complexity analysis

A standard 3×3 convolution over C channels costs $9C^2HW$ multiply–accumulates, whereas one DIDConv branch set costs $\left(k^2+2M\right)CHW$ for the three depthwise branches plus C^2HW for the 1×1 projection, so the depthwise terms scale linearly rather than quadratically in C. Combined with the slimmer channel allocation (64–384 versus ResNet-18′s 64–512), the backbone parameter count drops from 11.2 M to 4.5 M. At the whole-model level, this corresponds to the 33.5% parameter reduction of the C2f-DIMB-only variant in the ablation study (from19.88 M to 13.21 M). The complete three-module BR-DETR reaches 13.54 M parameters, 31.9% below the baseline. For the $N = H \times W$ tokens of the P_5 map, vanilla MHSA costs $\mathcal{O}(N^2C)$, whereas the Hi-Fi branch costs $\mathcal{O}(Nw^2(1-\alpha)C)$, linear in N for fixed w, and the Lo-Fi branch costs $\mathcal{O}(N^2\alpha C/s^2)$; with the defaults $\alpha = 0.5$, $w = 2$, $s = 2$, the attention cost at this layer falls to 31% of vanilla MHSA, consistent with the throughput gain of the AIFI-HiLo-only variant in Table 12.

Table 1
Dataset Distribution across Training, Validation, and Test Sets.

Defect Type	Total Instances	Training (70%)	Validation (20%)	Test (10%)
Crack	8249	5710	1656	883
Spallation	5418	3838	1051	529
Efflorescence	5383	3761	1104	518
Exposed Bars	2368	1705	434	229
CorrosionStain	2097	1537	379	181
Total	23,515	16,551	4624	2340

4. Experiments

4.1. Dataset description

A multi-source concrete bridge defect dataset was constructed from three public benchmarks, totaling 8,924 images across five categories: Crack, Spallation, Efflorescence, Exposed bars, and Corrosion Stain. All images containing the target categories were retained from CODEBRIM [49] (30 bridges, varying weather), 858 images were selected from SDNET2018 [50] (54 bridges, diverse lighting, shadows, and surface inclusions), and 6,806 images were sourced from a Roboflow bridge-defect dataset [51]. Because merging public sources risks duplicates, inconsistent annotations, and domain bias, three quality-control procedures were applied before splitting: near-duplicate images were removed by perceptual-hash filtering; semantically equivalent labels from different sources were unified after visual inspection; and 10% of images were independently re-annotated by a second engineer. The residual inter-source domain bias is quantified by leave-one-source-out validation. All annotations were converted to a unified normalized YOLO format under a consistent five-category taxonomy.

The dataset was partitioned into training, validation, and test sets at a 7:2:1 ratio with per-category stratified sampling; the instance counts are summarized in Table 1.

The scale distribution of all 23,515 instances was analyzed by the square root of the bounding-box area adapted from the MS COCO convention [52]. As illustrated in Fig. 8, instances are strongly concentrated at smaller scales: tiny (<32 px) and small (32–128 px) instances jointly account for 42.8% of the dataset (mean 228 px, median 149 px), with Crack and Efflorescence exhibiting the highest small-scale concentration (43.5% and 49.8% below 128 px) while Spallation skews larger (34.7% in the $\geq$ 256 px zone). These characteristics underline the necessity of a multi-scale detection architecture.

4.2. Data augmentation

Online augmentation is applied to every model during training, as illustrated in Fig. 9.HSV photometric transformations simulate illumination, weather, and shadow variation; geometric transformations (horizontal flipping, random scaling, small-angle rotation, translation) improve robustness to viewpoint, altitude-dependent scale, and oblique imaging; Mosaic increases contextual diversity and is disabled for the final 10 epochs to stabilize training; and MixUp smooths decision boundaries between visually similar categories. The identical pipeline is applied to every model evaluated in this paper, eliminating augmentation as a confounding factor.

4.3. Implementation details

All model development and training were conducted on a workstation equipped with an Intel Core i9-14900 K CPU, an NVIDIA GeForce RTX 4090 D GPU (24 GB GDDR6X), and 64 GB RAM, running Windows

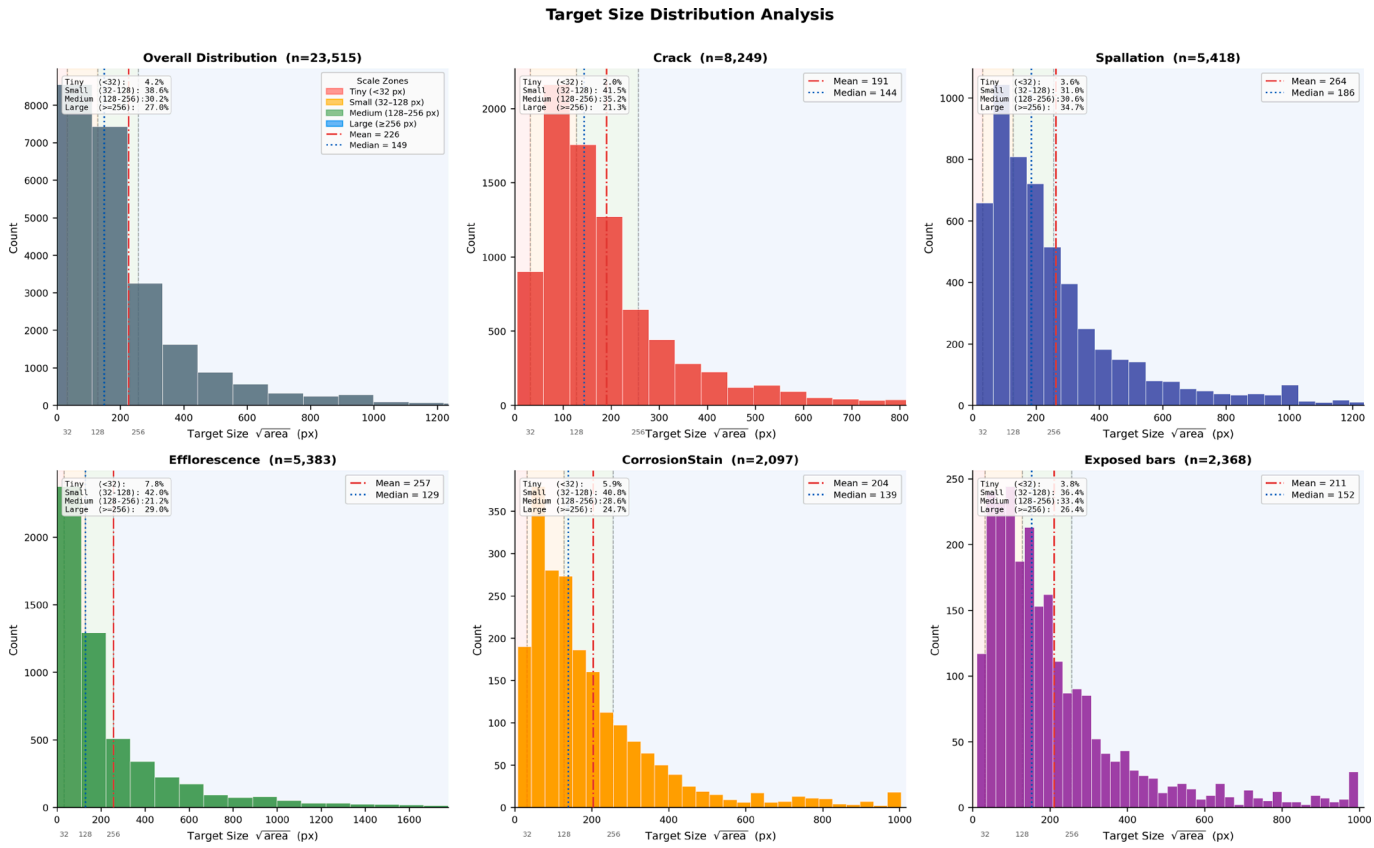


Fig. 8. Scale distribution of defect instances across five categories, quantified as $\sqrt{\text{area}}$(px). Scale zones: tiny (<32 px), small (32–128 px), medium (128–256 px), and large (≥256 px). Dashed and dotted vertical lines denote the mean and median.

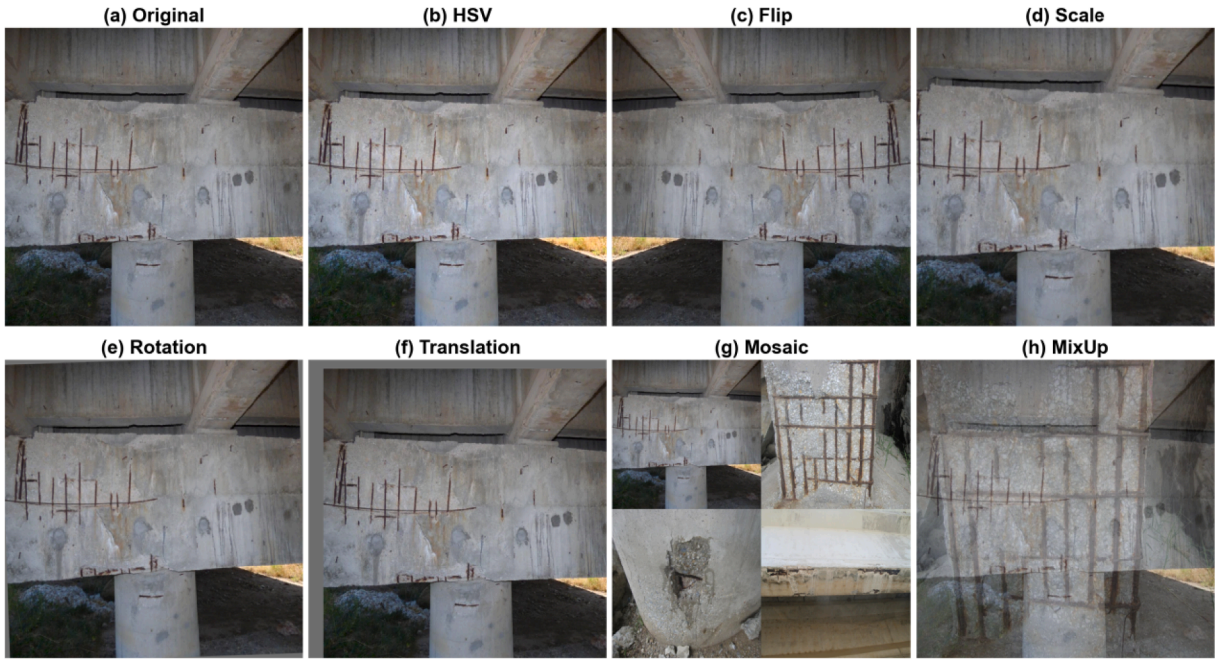


Fig. 9. Representative examples of online data augmentation strategies. (a) Original image; (b) HSV; (c) Horizontal flip; (d) Random scaling; (e) Random rotation; (f) Random translation; (g) Mosaic; (h) MixUp.

10 with PyTorch 2.3.0, CUDA 12.1, and Python 3.10.16. The trained models were subsequently deployed on the NVIDIA Jetson Xavier NX edge platform shown in Fig. 10 to validate real-time inference under the resource-constrained conditions representative of UAV-mounted systems. The detailed hardware specifications of the Jetson Xavier NX are listed in Table 2. The Xavier NX is integrated into a self-assembled quadrotor UAV shown in Fig. 11, with a 300 mm × 250 mm airframe and GPS + GLONASS dual-mode positioning. A CADDX Ratel 2 camera

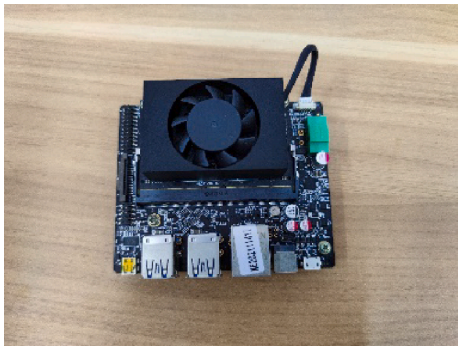


Fig. 10. The Jetson Xavier NX.

Table 2
The parameters of Jetson Xavier NX.

Parameter	The Jetson Xavier NX
GPU	384-core Volta GPU with 48 Tensor Cores
CPU	6-core Carmel ARM 64-bit CPU
Storage Capacity	16 GB EMMC + 128 GB SSD
Size	69.7 mm × 46 mm
AI Performance	21 TOPS
Memory	128-bit 16 GB LPDDR4x 59.7 GB/s
Power	10 W – 15 W

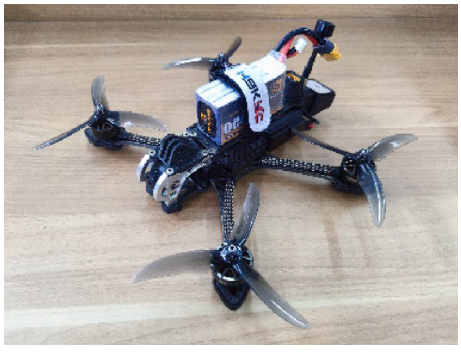


Fig. 11. The Unmanned aerial vehicle.

is wired directly to the Xavier NX so that frames are consumed onboard by the TensorRT engine, with results relayed to ground terminals over Wi-Fi. All embedded results in this paper were obtained on this platform.

Training protocol. A central question for a fair comparison is whether every model was trained with a well-tuned, near-optimal configuration rather than a single setting that happens to favor one architecture. Each model was therefore trained with the configuration recommended and tuned by its original authors, under which it reaches its designed performance shown in Table 3, the YOLO series and the classical CNN detectors (Faster R-CNN, SSD, RetinaNet [53]) use SGD with a batch size of 16, whereas the RT-DETR variants and BR-DETR use AdamW with a batch size of 4, the setting at which the DETR architecture converges best. Forcing a single common batch size would push some architectures away from their tuned optimum and would therefore be the less fair choice; the differing batch sizes reflect each model operating at its best. The remaining settings are shared: 300 epochs, a 640 × 640 input resolution, the identical online-augmentation pipeline, and the same 7:2:1 split.

4.4. Evaluation metrics

Model complexity and speed are quantified by parameters (M), FLOPs (G), and FPS. Detection accuracy is evaluated with precision, recall, F1-score, mAP50, and mAP50-95, following the standard COCO evaluation protocol. To assess geometric measurement quality, four metrics are computed over true-positive (TP) detections matched to ground truth at IoU ≥ 0.5 with category agreement. The Center Localization Error (CLE) is the mean Euclidean distance between predicted and ground-truth box centers:

$$\mathrm{CLE} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{(x_i-\widehat{x}_i)^2+(y_i-\widehat{y}_i)^2} \tag{15}$$

The Dimensional Estimation Error (DEE) is the mean relative error of the measured defect width and height:

$$\mathrm{DEE} = \frac{1}{2N}\sum_{i=1}^{N}\left(\frac{|w_i-\widehat{w}_i|}{w_i}+\frac{|h_i-\widehat{h}_i|}{h_i}\right)\times 100\% \tag{16}$$

The Area Quantification Error (AQE) is the mean relative error of the measured defect area:

$$\mathrm{AQE} = \frac{1}{N}\sum_{i=1}^{N}\frac{|A_i-\widehat{A}_i|}{A_i}\times 100\% \tag{17}$$

Finally, the strict-IoU ratios characterize the distribution of geometric uncertainty over the matched detections:

$$P_\tau = \frac{1}{N_{\mathrm{TP}}}\sum_{i=1}^{N_{\mathrm{TP}}}\mathbf{1}[\mathrm{IoU}_i\geqslant\tau], \qquad \tau\in\{0.75,\ 0.90\} \tag{18}$$

where N_{TP} is the number of matched TP detections and $\mathbf{1}[\cdot]$ is the indicator function. All four metrics are aggregate means over the matched TP detections of the test set. Since the public source datasets provide no camera-to-surface distance, camera intrinsics, or calibrated physical scale, localization and dimensional errors are reported in pixel units. A single illustrative GSD of 0.2 mm/pixel, corresponding to a representative UAV close-range working distance of about 1 m, is used only to translate pixel errors into physical units, not as a measured acquisition condition. These metrics link the detector output directly to the dimensional reliability required by bridge condition assessment.

Table 3
Hyperparameter configurations for all evaluated models.

Model	Learning Rate	Optimizer	Momentum	Weight Decay	Epochs	Batch Size
Faster R-CNN	1e-4	SGD	0.9	5e-4	300	16
SSD	6e-4	SGD	0.9	5e-4	300	16
RetinaNet	1e-2	SGD	0.937	5e-4	300	16
YOLOv5s	1e-2	SGD	0.937	5e-4	300	16
YOLOv8s	1e-2	SGD	0.937	5e-4	300	16
YOLOv12s	1e-2	SGD	0.937	5e-4	300	16
RT-DETR-R18	1e-4	AdamW	0.9	1e-4	300	4
RT-DETR-R34	1e-4	AdamW	0.9	1e-4	300	4
RT-DETR-R50	1e-4	AdamW	0.9	1e-4	300	4
BR-DETR (Ours)	1e-4	AdamW	0.9	1e-4	300	4

Table 4
Detection results of different models (input size: 640 × 640).

Method	mAP50 (%)	mAP50-95 (%)	Precision (%)	Recall (%)	F1-Score (%)	FPS	Params (M)	FLOPs (G)
Faster R-CNN	50.23	30.17	63.41	54.82	58.79	24	41.5	207.1
SSD	39.68	22.44	57.13	46.27	51.11	91	26.3	31.4
RetinaNet	54.37	34.51	67.82	57.94	62.52	41	36.4	97.8
YOLOv5s	56.43	37.82	67	58.64	62.68	**374**	**7.2**	**16.5**
YOLOv8s	63.85	43.27	72.46	63.17	67.49	261	11.2	28.8
YOLOv12s	65.83	45.18	74.63	64.52	69.21	254	9.4	21.5
RT-DETR-R18	67.84	48.69	82.37	60.53	68.91	298	19.88	57.0
RT-DETR-R34	69.53	51.28	83.12	62.44	71.34	226	31.11	88.8
RT-DETR-R50	70.87	52.74	83.64	63.18	72.03	153	41.96	129.6
BR-DETR	**72.51**	**54.36**	**84.47**	**65.61**	**73.41**	320	13.54	52.1

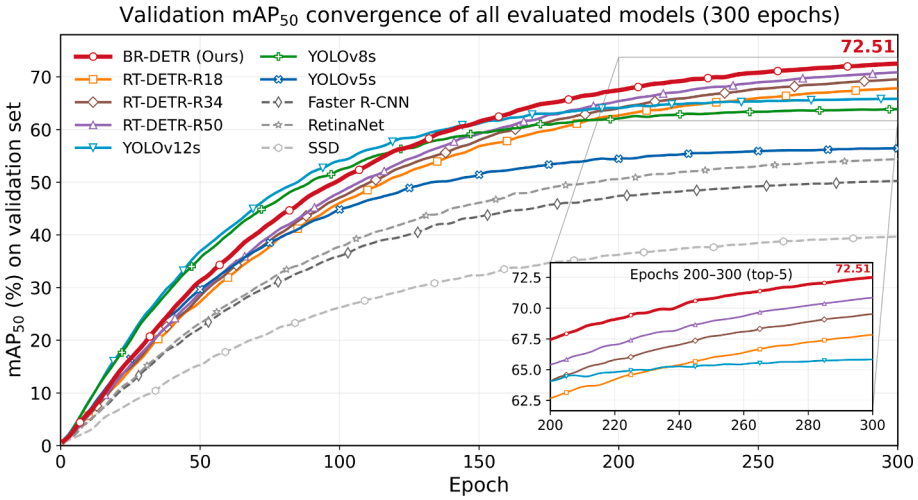


Fig. 12. Validation mAP50 convergence curves of all evaluated models.

5. Experimental results and analysis

5.1. Comparative experiments

BR-DETR is evaluated against representative methods spanning three paradigms (classical two-stage, one-stage CNN, and Transformer-based end-to-end detectors), each trained with its officially recommended configuration as described in Section 4.3. Inference speed is measured on a single RTX 4090 D GPU with 640 × 640 input and a batch size of 4. Results are presented in Table 4.

BR-DETR achieves the highest mAP50 and mAP50-95 among all evaluated methods, improving over RT-DETR-R18 by 4.67 and 5.67 pp while reducing parameters by 31.9% and GFLOPs by 8.6%, running at 320 FPS. Among classical detectors, Faster R-CNN is limited by its two-stage design, SSD's fixed anchors fail on the extreme scale variation of bridge defects, and RetinaNet remains too costly for the edge deployment. Within the YOLO family, even the most accurate YOLOv12s falls 6.68 pp below BR-DETR in mAP50, reflecting the limitation of pure CNN architectures in modelling the long-range dependencies needed to separate visually similar categories. Scaling RT-DETR to ResNet-50 raises accuracy but at substantially higher cost: with only one-third of the parameters and two-fifths of the GFLOPs, BR-DETR still surpasses RT-DETR-R50 in mAP50.

Training convergence. Fig. 12 plots the validation mAP50 of all ten models over the 300-epoch schedule. Every model plateaus well before the end of training, with CNN detectors typically within 150 epochs and Transformer detectors within 250, confirming that the reported results are not affected by under-training despite the different optimizers and batch sizes.

Unified-protocol control experiment. To directly address potential protocol-induced bias, the four strongest competitors and BR-DETR were retrained under one identical configuration (AdamW, lr = 1×10^{-4} with cosine schedule, batch 8, identical augmentation, 300 epochs). As shown in Table 5, the ranking is unchanged: BR-DETR keeps a 2.88 pp margin over RT-DETR-R34 and a 6.97 pp margin over the strongest YOLO variant, confirming that the conclusions of Table 4 are not artifacts of per-model protocol differences.

Table 5
Detection accuracy under each model's official configuration versus a unified training protocol (AdamW, lr = 1×10^{-4}, batch = 8, 300 epochs).

Model	mAP50 (%)official protocol	mAP50 (%)unified protocol
YOLOv8s	63.85	62.17
YOLOv12s	65.83	65.21
RT-DETR-R18	67.84	67.52
RT-DETR-R34	69.53	69.30
BR-DETR	72.51	72.18

Statistical stability. BR-DETR and RT-DETR-R18 were each trained with three random seeds, giving validation mAP50 of 72.48 ± 0.18% and 67.79 ± 0.23% (mean ± std); a two-sample *t*-test confirms that the 4.69 pp gap is significant ($p < 0.001$), and all single-run results reported in this work lie within one standard deviation of the seed means.

5.2. Embedded deployment evaluation on the Jetson Xavier NX

For UAV-based real-time inspection, a minimum of 30 FPS is adopted as the onboard requirement. All embedded results use the following stack: JetPack 5.1.2 (L4T 35.4.1), CUDA 11.4, TensorRT 8.5.2; models are exported to ONNX (opset 17) and built into FP16 engines with a 2 GB

Table 6
Detection speed and mAP50 of different models on the Jetson Xavier NX (TensorRT FP16, batch size 1, input size: 640 × 640).

Method	FPS	mAP50 (%)
Faster R-CNN	11	49.87
SSD	20	39.42
RetinaNet	16	54.05
YOLOv5s	44	56.21
YOLOv8s	38	63.58
YOLOv12s	37	65.49
RT-DETR-R18	40	67.39
RT-DETR-R34	31	69.05
RT-DETR-R50	21	70.32
BR-DETR (Ours)	**46**	**72.08**

workspace, no INT8 calibration. Inference runs at batch size 1 with a 50-frame warm-up; reported latencies cover the full on-device pipeline including pre-processing and post-processing. The Xavier NX runs in the 15 W six-core mode throughout. FP16 quantization incurs approximately 0.4 pp mAP50 degradation for either model. The detection speed and accuracy of all evaluated models are listed in Table 6.

As shown in Table 6, the three classical detectors (Faster R-CNN, SSD, RetinaNet) fall well below the 30 FPS threshold at 11–20 FPS, and RT-DETR-R50 also fails to meet the requirement at 21 FPS. Among the compliant models, BR-DETR achieves the highest mAP50 of 72.08% at 46 FPS, providing the best accuracy–throughput balance in this group.

A single FPS figure, however, does not capture the latency variability that governs worst-case behavior on a moving platform. To characterize this, the per-frame latency of BR-DETR was profiled over 1,000 consecutive frames; the resulting distribution is shown in Fig. 13(a), with key statistics reported in Table 7. BR-DETR achieves a mean latency of 21.7 ± 1.2 ms and a 99th-percentile latency of 25.1 ms, both well within the 33.3 ms budget. Compared with the RT-DETR-R18 baseline (mean: 25.0 ms, P99: 29.0 ms), BR-DETR not only reduces mean latency but also tightens the tail of the distribution, which is critical for UAV use where worst-case latency determines whether a frame captured in motion is processed before the next acquisition. On the resource side, BR-DETR reduces runtime memory by 18.5% (from 856 MB to 698 MB) and average board power by 1.2 W; at 11.2 W, the inference payload

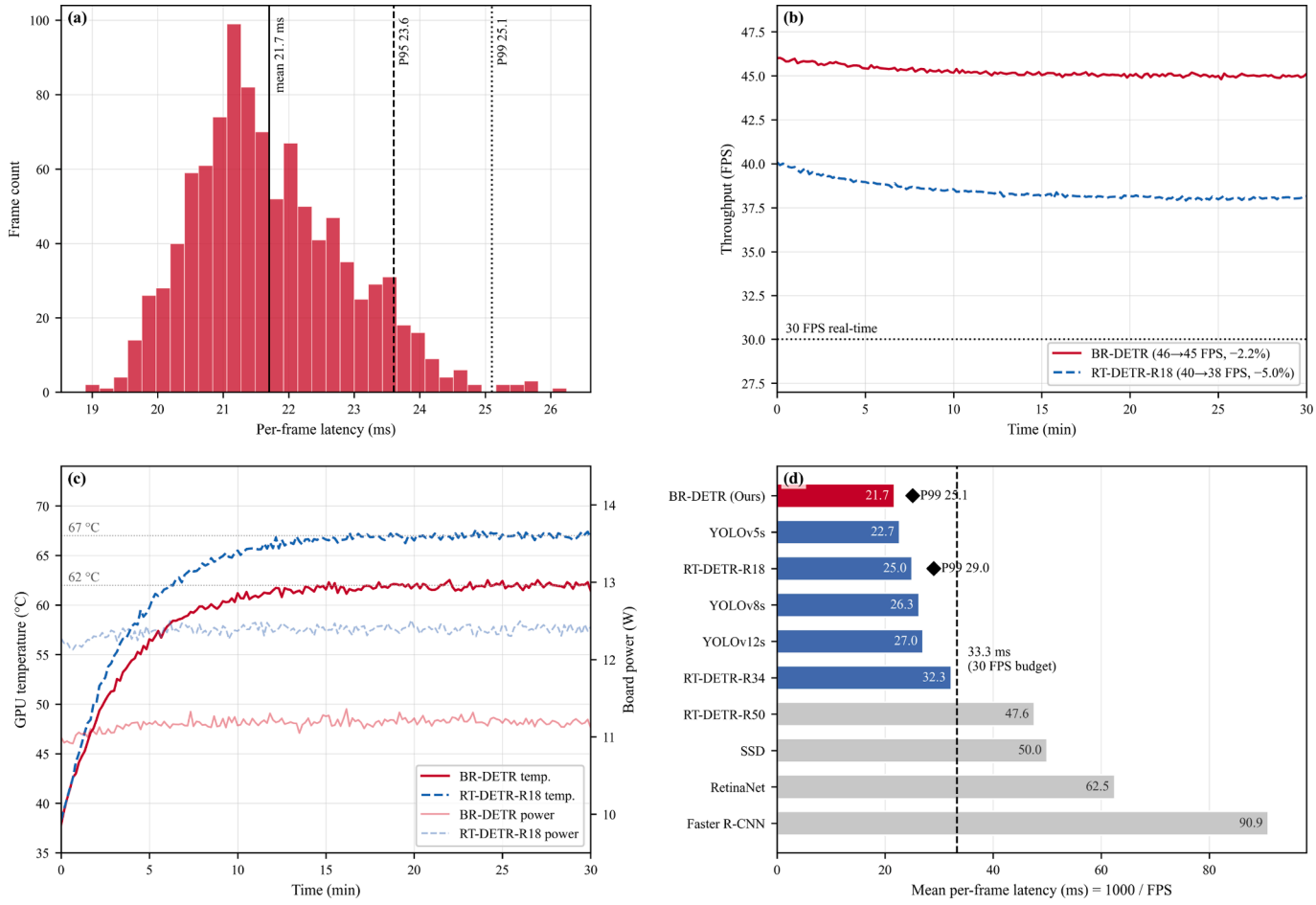


Fig. 13. Embedded deployment evaluation on the Jetson Xavier NX: (a) per-frame latency distribution of BR-DETR over 1,000 frames with mean, P95, and P99 marked; (b) sustained throughput over the 30-minute test; (c) GPU temperature and board power draw over the same window; (d) cross-model mean latency with 33.3 ms real-time threshold.

Table 7
Per-frame latency statistics and runtime resource footprint on the Jetson Xavier NX (TensorRT FP16, batch = 1, 1,000 frames, 15 W mode).

Model	Mean (ms)	Std (ms)	P95 (ms)	P99 (ms)	Equivalent FPS	GPU util. (%)	Runtime memory (MB)	Avg. power (W)	FP16 engine (MB)
RT-DETR-R18	25.0	1.4	27.2	29.0	40	83	856	12.4	41.2
BR-DETR	21.7	1.2	23.6	25.1	46	76	698	11.2	28.6

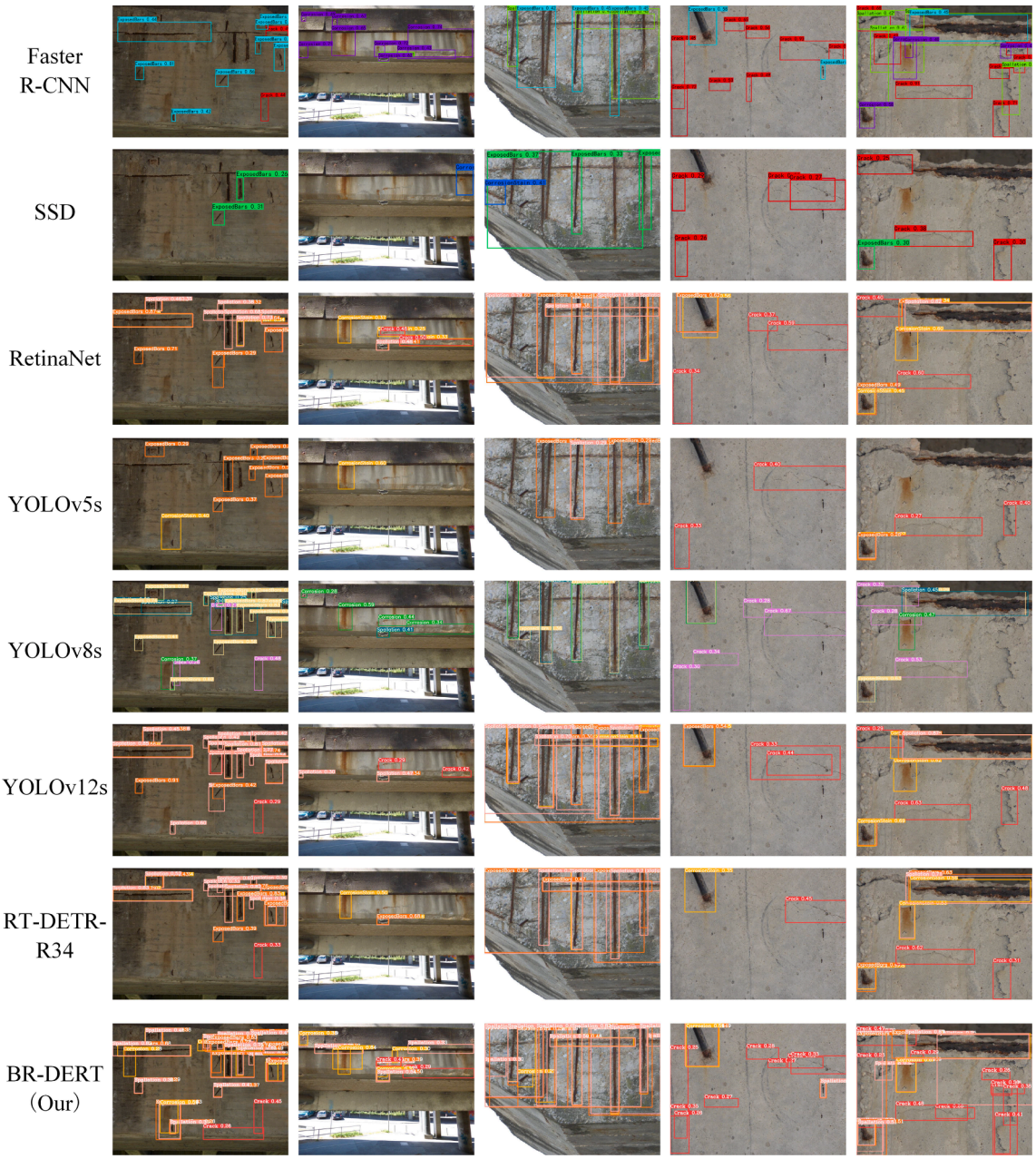


Fig. 14. Detection results comparison with different methods.

Table 8
Qualitative summary of the detection-visualization comparison.

Model	Missed Detection	False Detection	Confidence Level	Key Limitation
Faster R-CNN	Yes(CorrosionStain, Crack)	Yes(Efflorescence)	Low	Slow; missessmall defects
SSD	Yes(Crack, CorrosionStain, Efflorescence)	No	Low	Severe under-detection;
RetinaNet	Yes(CorrosionStain)	Yes(Efflorescence)	Moderate	Redundant bounding boxes; category confusion under complex backgrounds
YOLOv5s	Yes (CorrosionStain, Exposed bars, Efflorescence)	No	Moderate	Low-contrast defect omission
YOLOv8s	Yes(Efflorescence)	Yes(CorrosionStain)	Moderate	Incomplete detection in dense defect regions
YOLOv12s	Yes (CorrosionStain)	Yes(Efflorescence)	Moderate	Category confusion
RT-DETR-R34	Yes (CorrosionStain)	Yes(CorrosionStain)	Moderate	Uniform attention limits discrimination of low-contrast defects
BR-DETR (Ours)	**No** (in the visualized samples)	**No** (in the visualized samples)	High (>0.8)	–

consumes approximately 9% less battery energy per sortie, directly benefiting flight endurance.

Table 9
Class-specific Performance comparison between baseline and BR-DETR on the validation split.

Defect Type	Method	Precision (%)	Recall (%)	F1-Score (%)	mAP50 (%)	mAP50-95 (%)
Corrosion Stain	Baseline	69.86	32.89	45.28	40.49	26.86
	BR-DETR	72.62	52.22	60.75	49.56	31.51
		+2.76	+19.33	+15.47	+9.07	+4.65
Crack	Baseline	89.57	61.70	73.07	69.61	47.30
	BR-DETR	92.57	65.42	76.66	72.87	54.57
		+3.00	+3.72	+3.59	+3.26	+7.27
Efflorescence	Baseline	80.63	77.32	78.94	82.44	60.49
	BR-DETR	86.99	84.69	85.82	88.99	70.14
		+6.36	+7.37	+6.88	+6.55	+9.65
Exposed Bars	Baseline	82.84	59.86	69.58	68.66	46.85
	BR-DETR	83.06	61.17	70.37	70.62	49.43
		+0.22	+1.31	+0.79	+1.96	+2.58
Spallation	Baseline	85.95	70.88	77.69	78.00	61.95
	BR-DETR	90.11	74.53	81.58	80.50	66.15
		+4.16	+3.65	+3.89	+2.50	+4.20

Table 10
Per-category mAP50 (%) comparison across representative models on the validation split.

Model	Crack	Spallation	Efflorescence	Exposed Bars	Corrosion Stain	All
YOLOv8s	65.25	73.85	77.68	64.12	38.71	63.85
YOLOv12s	67.43	75.96	79.89	66.50	39.67	65.83
RT-DETR-R18	69.61	78.10	82.44	68.66	40.49	67.84
RT-DETR-R34	70.85	79.23	84.32	69.70	43.74	69.53
BR-DETR	**72.87**	**80.50**	**88.99**	**70.62**	**49.56**	**72.51**

Thermal behavior was assessed by a 30-minute sustained-inference test at 25 °C ambient with the stock active cooler; the throughput and thermal traces are shown in Fig. 13(b) and (c), respectively. BR-DETR stabilizes at approximately 62 °C after 12 min with a throughput loss of only 2.2% (46 to 45 FPS), versus 67 °C and 5.0% (40 to 38 FPS) for the baseline. Neither model triggers thermal throttling within the test window.

Beyond the deployment metrics reported above, qualitative inspection of representative outputs provides complementary insight into model behaviour. As illustrated in Fig. 14 and summarized in Table 8, the comparison models exhibit visibly more missed and false detections than BR-DETR on the displayed samples, particularly on the low-contrast Corrosion Stain and Efflorescence categories. This observation is necessarily limited to the visualized samples and does not contradict the dataset-level recall of 65.61%, which reflects the full test set including heavily occluded and extremely low-contrast instances.

5.3. Class-specific and scale-specific performance analysis

Detailed per-category results are presented in Table 9. BR-DETR improves on the baseline across all five categories, with gains that reflect individual defect characteristics. Corrosion Stain records the largest absolute improvement of 9.07 pp, validating the fine-grained spatial information preserved by MSDEP for small-scale defects. Crack and Spallation gain 3.26 and 2.50 pp, respectively, benefiting from the adaptive multi-branch convolutions of C2f-DIMB. Efflorescence reaches the highest category mAP50 of 88.99% (+6.55 pp), indicating that AIFI-HiLo strengthens discriminative learning for low-contrast defects requiring global context. Exposed Bars yields the most modest gain of 1.96 pp, suggesting persistent rebar–crack visual similarity.

On the held-out test split, BR-DETR attains 71.96% mAP50, 53.25% mAP50-95, and a 72.73% F1-score, confirming that the validation performance generalizes to unseen data. The relative category difficulty is preserved, with Corrosion Stain remaining the most challenging category. This held-out split is used for all measurement and robustness analyses in the remainder of this section.

To verify that the advantage of BR-DETR is targeted rather than uniform, Table 10 extends the per-category comparison to the strongest CNN and Transformer baselines. BR-DETR leads on every category, but the margins are largest on the two measurement-critical categories, Corrosion Stain (5.82 pp over the best baseline) and Efflorescence (4.67 pp), while narrowing to 1.27 pp on the geometrically distinctive Spallation.

Table 11
Detection performance (mAP50, %) by object size, small ($\sqrt{\text{area}} < 128$ px), medium (128–256 px), large (≥ 256 px).

Model	Small (%)	Medium (%)	Large (%)
YOLOv8s	33.52	58.73	71.22
YOLOv12s	35.43	60.47	73.04
RT-DETR-R18 (Baseline)	38.27	62.59	75.38
RT-DETR-R34	40.10	64.21	76.85
BR-DETR	**46.70**	**65.82**	**79.13**

Scale-specific comparison. Table 11 reports detection performance stratified by object size. Following the scale distribution analysis in Section 4.1, which showed that 42.8% of instances fall below 128 px, we merge the tiny and small bins into a single "small" category and evaluate at three thresholds: small, medium, and large. BR-DETR achieves its largest gain on small objects (+8.43 pp over the baseline, a 22.0% relative improvement), confirming that MSDEP preserves the fine-grained spatial information for small-scale defect detection. Improvements diminish to 3.23 and 3.75 pp for medium and large objects, respectively. The small-object margin over RT-DETR-R34 (6.6 pp) is more than twice the overall margin (2.98 pp), further confirming that MSDEP's benefit is concentrated at the small-object scale.

5.4. Ablation study

Ablation experiments on the validation and test splits evaluate each module individually, all pairwise combinations, and the full model; results are shown in Table 12.

Each module yields a consistent and interpretable improvement on both splits. MSDEP provides the largest individual accuracy gain (+4.45

Table 12
Results of the BR-DETR ablation experiment.

Data	RT-DETR-R18	C2f-DIMB	MSDEP	AIFI-HiLo	mAP50 (%)	mAP50-95 (%)	F1-Score (%)	Params (M)	FLOPs (G)	FPS
Val	√	–	–	–	67.84	48.69	68.91	19.88	57.0	298
	√	√	–	–	69.35	53.44	70.64	13.21	**43.6**	**364**
	√	–	√	–	72.29	51.83	73.22	20.01	62.4	278
	√	–	–	√	70.50	50.53	70.51	19.84	57.1	309
	√	√	√	–	71.47	52.18	71.90	13.58	51.9	294
	√	√	–	√	70.86	53.92	71.18	**13.17**	43.7	348
	√	–	√	√	72.38	53.07	73.05	19.97	62.5	252
	√	√	√	√	**72.51**	**54.36**	**73.41**	13.54	52.1	320
Test	√	–	–	–	66.52	47.90	66.55	19.88	57.0	298
	√	√	–	–	68.31	53.18	69.55	13.21	**43.6**	**364**
	√	–	√	–	71.67	50.83	72.26	20.01	62.4	278
	√	–	–	√	69.84	48.53	69.47	19.84	57.1	309
	√	√	√	–	70.92	52.83	71.23	13.58	51.9	294
	√	√	–	√	70.12	52.85	70.43	**13.17**	43.7	348
	√	–	√	√	71.80	52.10	72.31	19.97	62.5	252
	√	√	√	√	**71.96**	**53.25**	**72.73**	13.54	52.1	320

Table 13
Measurement performance comparison between the baseline and BR-DETR on the test set (↓: lower is better; ↑: higher is better).

Metric	RT-DETR−R18	BR-DETR	Change
Center Localization Error, CLE (px) ↓	6.83	4.92	−28.0%
CLE in physical units (mm @ GSD 0.2 mm/px) ↓	1.37	0.98	−28.0%
Dimensional Estimation Error, DEE (%) ↓	12.4	9.2	−3.2 pp
Area Quantification Error, AQE (%) ↓	18.7	13.6	−5.1 pp
Share of TP with IoU ≥ 0.75, $P_{0.75}$ (%) ↑	52.3	61.8	+9.5 pp
Share of TP with IoU ≥ 0.90, $P_{0.90}$ (%) ↑	18.5	24.7	+6.2 pp

pp mAP50 on the validation split), validating its fine-grained small-object feature pathway. C2f-DIMB contributes + 1.51 pp while delivering the largest complexity reduction (33.5% fewer parameters and 23.5% fewer GFLOPs). AIFI-HiLo adds + 2.66 pp through frequency-aware attention with no parameter overhead. Two pairwise interactions merit further discussion. First, C2f-DIMB + MSDEP (71.47%) falls slightly below MSDEP alone (72.29%), as the aggressive parameter reduction of the lightweight backbone marginally limits the capacity that the enriched feature pyramid can exploit. Second, the full combination (72.51%) surpasses all pairwise results, because AIFI-HiLo recovers the boundary–context separation that the slimmer backbone alone cannot provide; MSDEP + AIFI-HiLo (72.38%) approaches this accuracy but requires 47.5% more parameters (19.97 M vs. 13.54 M) at only 252 FPS. Regarding throughput, MSDEP alone reduces FPS from 298 to 278, while C2f-DIMB's efficiency gain progressively offsets this overhead across pairwise and full-model configurations, with the complete model reaching 320 FPS. The full three-module combination is therefore the only configuration that attains the highest accuracy (72.51% mAP50) while remaining within the lightweight budget (13.54 M, 31.9% below the baseline); the other high-accuracy variants require roughly 20 M parameters, and the other lightweight variants sacrifice 1–3 pp of accuracy. This provides quantitative evidence that the three modules are complementary along the spatial, resolution, and frequency axes.

5.5. Measurement performance analysis

This section evaluates BR-DETR as a measurement instrument using the metrics of Section 4.4, computed over TP detections matched at IoU ≥ 0.5 with category agreement on the test set. Because the source datasets provide no camera-to-surface distance or camera calibration, all errors are reported in pixels; a single illustrative GSD of 0.2 mm/pixel is used only to express the center localization error in physical units. Results are presented in Table 13, with per-class breakdowns shown in Fig. 15.

BR-DETR reduces the mean center localization error from 6.83 px to 4.92 px (−28.0%; approximately 0.98 mm at the illustrative GSD), confirming that the P2-level feature injection of MSDEP translates into tangible geometric gains rather than only a higher mAP. The dimensional and area errors, which directly govern crack-width, crack-length, and spallation-area quantification, decrease from 12.4% to 9.2% (−3.2 pp) and from 18.7% to 13.6% (−5.1 pp), respectively. The rightward shift of the IoU distribution — P0.75 rising from 52.3% to 61.8% and P0.90 from 18.5% to 24.7% — indicates a systematic reduction in

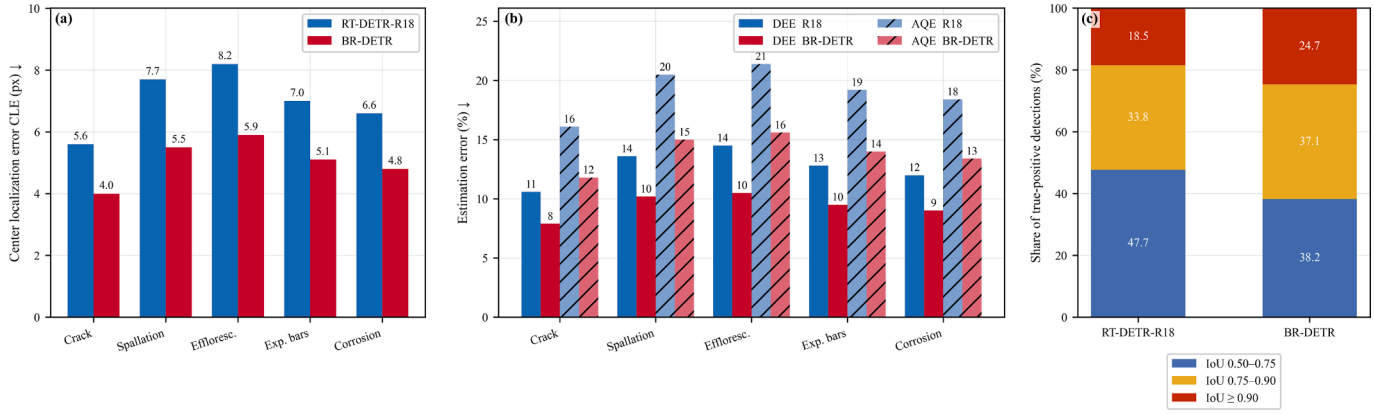


Fig. 15. Measurement performance on the test set: (a) per-class center localization error; (b) per-class dimensional and area estimation errors; (c) IoU-band distribution of true-positive detections.

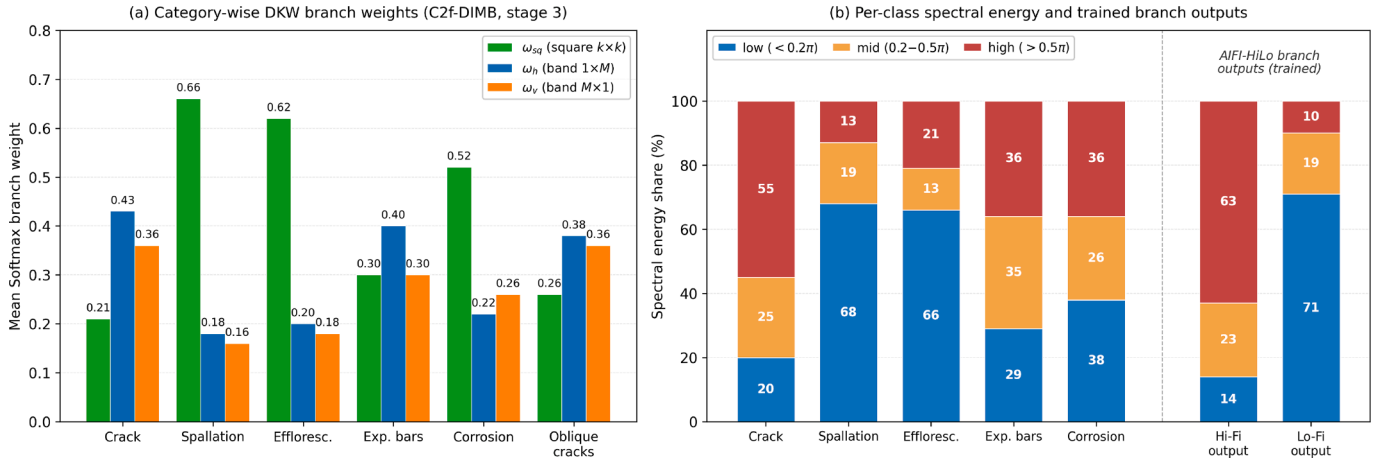


Fig. 16. Mechanism-level evidence for C2f-DIMB and AIFI-HiLo: (a) category-wise mean DKW branch weights at stage 3 (50 images per category); (b) per-class spectral energy distribution of ground-truth defect patches and of the trained Hi-Fi and Lo-Fi branches.

Table 14
Comparison of MSDEP against alternative small-object enhancement strategies, all built on the same RT-DETR-R18 baseline. (input size: 640 × 640).

Scheme	mAP50 Small (%)	mAP50 All (%)	Params (M)	FLOPs (G)	FPS (Jetson, TRT FP16)
Baseline (P_3–P_5)	38.27	67.84	19.88	57.0	40
+ P_2 detection head	45.85	70.93	21.60	78.2	**27**
+ BiFPN [39]	41.37	69.21	20.31	64.3	35
+ HS-FPN [44]	40.62	68.75	19.64	59.4	38
+ **MSDEP**	**45.91**	**72.29**	20.01	62.4	36

geometric uncertainty, with a larger share of detections reaching the accuracy required for severity grading.

The residual missed-detection rate of 34.4% (the complement of the 65.61% test-set recall) represents a known limitation that should be acknowledged for operational use. In its current form, BR-DETR is suited to early-warning screening workflows, while safety–critical structures may still require multi-pass scanning or human review of low-confidence regions. As shown in Table 9, per-category recall is substantially higher for the deterioration-indicative Spallation (74.5%) and Efflorescence (84.7%) categories than for early-stage Corrosion Stain (52.2%), indicating that the categories with the greatest immediate maintenance consequence are the least likely to be missed — the preferable failure mode for an automated inspection instrument.

5.6. Comparison with alternative Small-Object enhancement strategies

To validate the MSDEP design, it is compared against three representative small-object enhancement strategies built on the identical RT-DETR-R18 baseline with backbone and encoder unchanged: (i) a genuine P_2 detection head on the standard CCFF; (ii) BiFPN [39]; and (iii) HS-FPN [44]. Table 14 reports accuracy and deployment-side costs for all four schemes.

The trade-off space can be characterized as follows. A P2 head improves small-object mAP50 by 7.6 pp, but introduces 1.72 M additional parameters and 21.2 GFLOPs; the additional P2 feature map reduces Jetson throughput to 27 FPS, falling below the real-time threshold required for onboard deployment. Generic enhancements (BiFPN, HS-FPN) preserve inference speed but recover less than half of the small-object gain (+3.1 and + 2.4 pp, respectively). MSDEP attains a comparable small-object improvement (+7.6 pp) and the highest overall mAP50 in this group (72.29%), while adding only 0.13 M parameters and 5.4 GFLOPs and maintaining 36 FPS. This accuracy–efficiency operating point is not matched by any of the alternative strategies evaluated, supporting the practical suitability of MSDEP for UAV-based deployment.

5.7. Mechanism analysis: dynamic kernel weights and frequency decomposition

To examine whether the two adaptive designs operate as intended, this section analyzes the learned DKW behavior of C2f-DIMB and the frequency composition underlying AIFI-HiLo.

Dynamic kernel weights. For each category, 50 single-category test images were forwarded through the trained model, and the Softmax-normalized branch weights $\boldsymbol{\omega} = [\omega_{sq}, \omega_h, \omega_v]$ of the stage-3 DIDConv modules were averaged. Fig. 16 (a) summarizes the results, which are consistent with the morphological prior. The Crack category assigns the majority of its weight to the band branches ($\omega_h = 0.43$, $\omega_v = 0.36$, $\omega_{sq} = 0.21$; combined band-branch share 0.79), while Spallation and Efflorescence strongly favor the isotropic branch ($\omega_{sq} = 0.66$ and 0.62, respectively). Corrosion Stain occupies an intermediate position ($\omega_{sq} = 0.52$), consistent with its area-type yet textured appearance. Exposed bars direct 0.70 of its total weight to the band branches, with the dominant direction aligned with the prevailing bar orientation in the dataset. Notably, on a curated subset of obliquely oriented cracks, the two band branches are activated jointly and almost equally ($\omega_h = 0.38$, $\omega_v = 0.36$, $\omega_{sq} = 0.26$). This empirically confirms that oblique structures are represented by a convex combination of the two band branches rather than requiring a dedicated kernel per orientation.

Frequency composition. Fig.16(b) reports the per-class spectral energy distribution obtained from 2D-FFT analysis of ground-truth patches, partitioned into low ($< 0.2\pi$), mid (0.2–0.5π), and high ($> 0.5\pi$) normalized-frequency bands. The five categories exhibit the high–low frequency dichotomy assumed in Section 3.4: the Crack category concentrates 55% of its energy in the high band, whereas Spallation and Efflorescence concentrate 68% and 66% in the low band, respectively. Corrosion Stain and Exposed Bars occupy intermediate spectral profiles (low/mid/high of 38%/26%/36% and 29%/35%/36%, respectively). Applying the same analysis to the branch outputs of the trained encoder shows the Hi-Fi branch concentrating 63% of its output energy in the high band and the Lo-Fi branch 71% in the low band, confirming that the intended spectral specialization is realized through end-to-end training. Qualitative attention maps corroborate this finding: Hi-Fi attention localizes defect boundaries, while Lo-Fi attention spreads over the surrounding structural context.

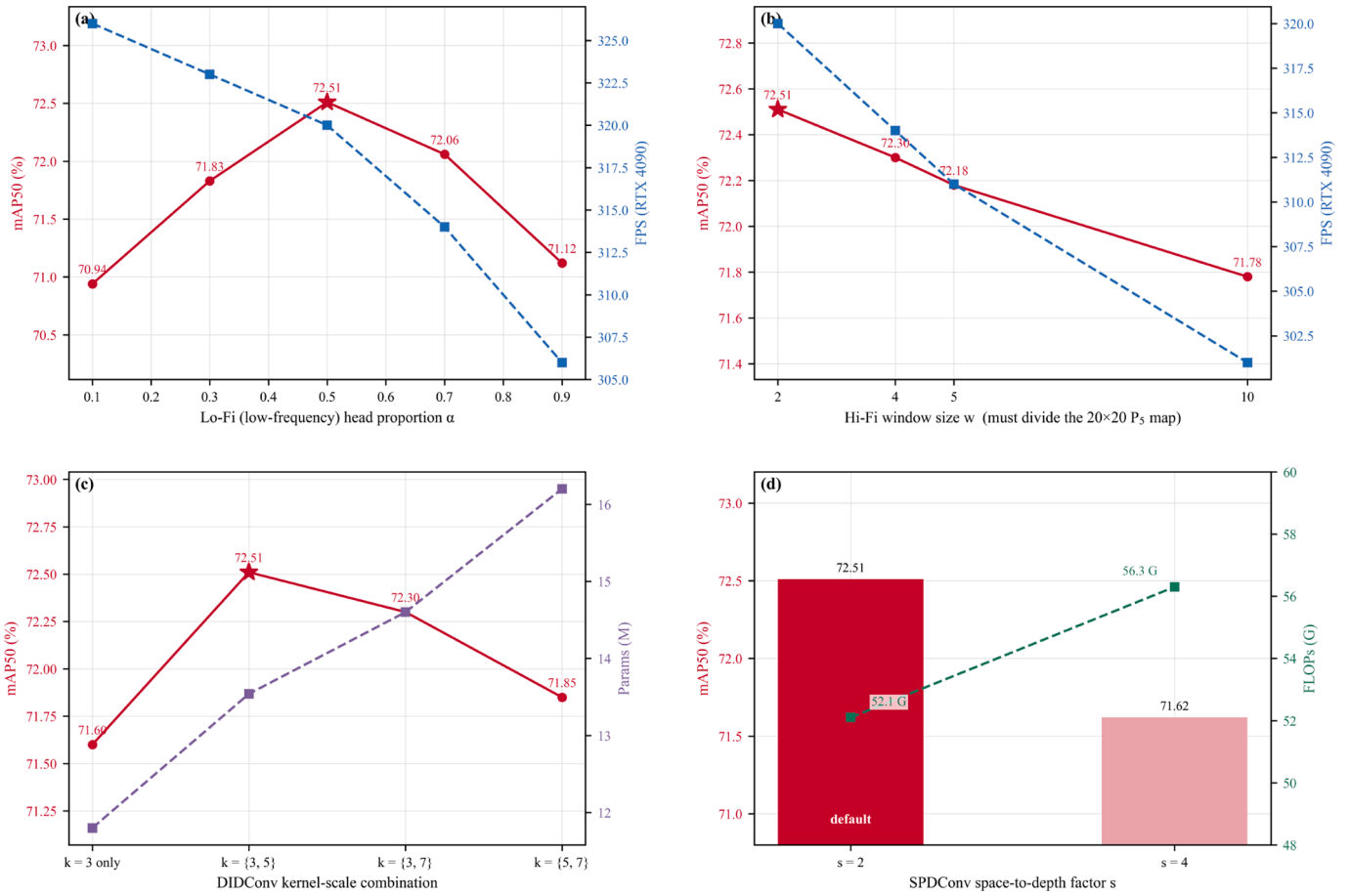


Fig. 17. Hyperparameter sensitivity of BR-DETR on the validation split: (a) HiLo head-split ratio α;(b) Hi-Fi window size w; (c)DIDConv kernel-scale combination;(d) SPDConv factor s.

5.8. Hyperparameter sensitivity analysis

Four critical hyperparameters are varied while all other settings are fixed at their defaults: the HiLo head-split ratio α (the proportion of Lo-Fi heads), the Hi-Fi window size w, the DIDConv kernel-scale combination, and the SPDConv space-to-depth factor s. Because w must evenly divide the 20 × 20P_5 token map, its admissible values are restricted to w ∈ {2, 4, 5, 10}, with w = 2 as the default. Validation results are plotted in Fig. 17.

The head-split ratio α has the largest influence on accuracy, with mAP50 varying by 1.57 pp across α ∈ [0.1, 0.9] (70.94–72.51%). Both extremes are inferior: too few Lo-Fi heads weaken global-context aggregation, while excessive Lo-Fi allocation insufficiently preserves high-frequency boundaries. The symmetric split α = 0.5 is optimal, reaching 72.51% mAP50 at 320 FPS, indicating that the choice of α is interpretable from the spectral characteristics of the target defects rather than arbitrary.

The SPDConv factor is the second most influential, with a 0.89 pp drop at s = 4 caused by excessive loss of spatial granularity in a single downsampling step. The window size w and kernel-scale combination k affect accuracy by less than the SPDConv factor, indicating that the model is robust to these secondary choices, with the defaults (w = 2, k = {3, 5}) at their respective optima.

5.9. Cross-dataset generalization and robustness analysis

Cross-dataset validation. Because the dataset merges three public sources, a leave-one-source-out (LOSO) protocol quantifies inter-dataset domain bias: in each fold the model is trained on two sources and tested on the entire held-out source. Table 15 reports the results.

Table 15
Leave-one-source-out cross-dataset validation (mAP50, %).

Training sources	Held-out test source	RT-DETR-R18	BR-DETR
CODEBRIM + Roboflow	SDNET2018	59.72	62.35 (+2.63)
SDNET2018 + Roboflow	CODEBRIM	60.54	64.21 (+3.67)
CODEBRIM + SDNET2018	Roboflow	63.27	67.43 (+4.16)
Combined (in-domain reference)	Combined test split	67.84	72.51 (+4.67)

As expected for domain shifts of this magnitude, absolute accuracy drops by 5–10p p when an entire source is withheld, the largest drop occurring on the texture-dominated SDNET2018 fold. However, BR-DETR consistently outperforms the baseline by 2.6–4.2 pp across all three folds, approaching its in-domain advantage of 4.67 pp, which indicates that the proposed modules improve transferable feature representations. A per-source breakdown of the in-domain test set further confirms balanced accuracy across origins (73.8% for CODEBRIM, 68.5% for SDNET2018, and 72.2% for Roboflow), all within 4 pp of the overall figure, confirming that no single source disproportionately influences the reported accuracy.

Perturbation robustness. To characterize model robustness under degraded acquisition conditions, the test set is corrupted offline at increasing severity levels: linear motion blur up to 20 px (simulating platform vibration), additive Gaussian noise up to σ = 30, and JPEG

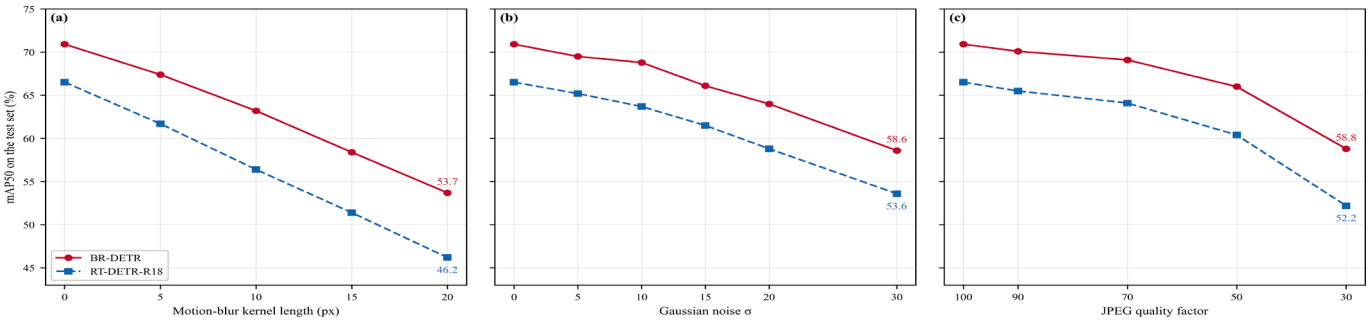


Fig. 18. Robustness to acquisition perturbations on the test set: (a) motion blur; (b) Gaussian noise; (c) JPEG compression.

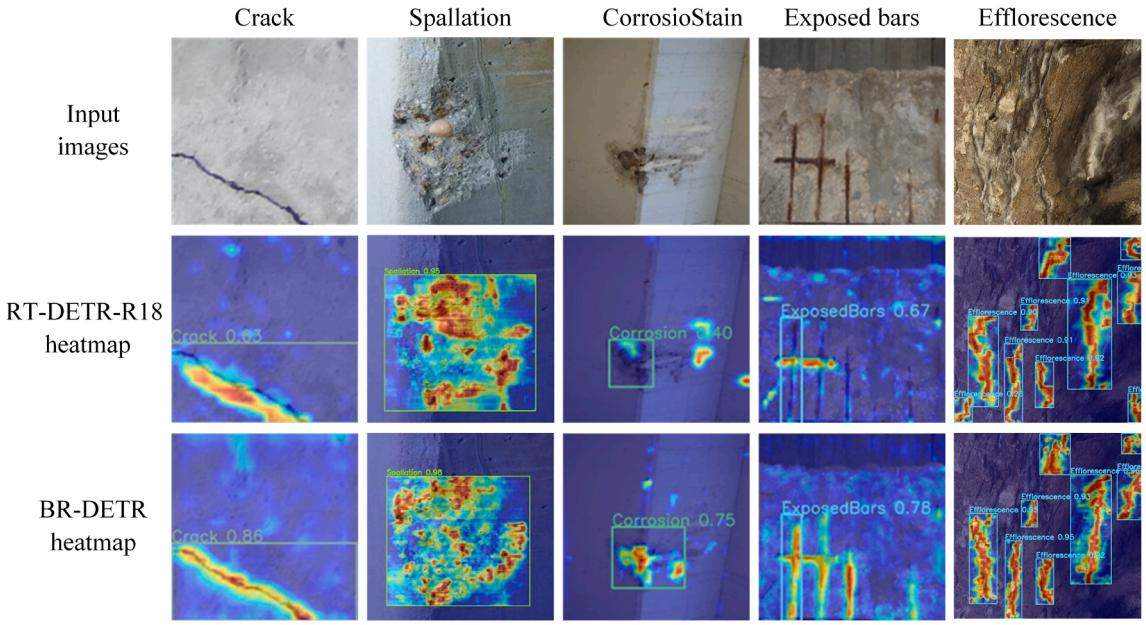


Fig. 19. Comparison of feature visualization heatmaps between RT-DETR-r18 and BR-DETR.

compression down to quality 30 (simulating bandwidth-limited transmission). Both models are re-evaluated on the corrupted data without retraining; results are shown in Fig. 18. BR-DETR exhibits greater robustness under all three corruption types. Under 15 px motion blur, it retains 58.4% mAP50 versus 51.4% for the baseline, widening the clean-image gap from 4.4 to 7.0 pp, consistent with the low-pass characteristic of the Lo-Fi branch. Under $\sigma = 30$ Gaussian noise, the retained accuracy is 58.6% versus 53.6%, and under quality-30 JPEG compression it is 58.8% versus 52.2%. Together with the LOSO results, these experiments provide quantitative evidence that the advantage of BR-DETR over the baseline persists under both domain shift and acquisition degradation, and widens under several corruption conditions.

5.10. *Heatmap visualization and analysis*

Grad-CAM [54] visualizations in Fig. 19 compare the feature-activation distributions of RT-DETR-R18 and BR-DETR across the five defect categories. For Crack, the baseline distributes attention broadly over intact concrete regions, whereas BR-DETR concentrates activation along crack patterns regardless of shape and scale, consistent with the adaptive receptive-field geometry of C2f-DIMB. For Spallation, the baseline produces diffuse activation that fails to cover the full extent of the damaged region, while BR-DETR generates more extensive and coherent coverage, attributed to the enrichment of local features with global context by MSDEP. For Efflorescence, the baseline's scattered activation fails to separate defects from normal surface variation, while BR-DETR produces concentrated activation reflecting the frequency-aware discrimination of AIFI-HiLo. For the two lower-recall categories, Exposed Bars and Corrosion Stain, BR-DETR produces more spatially precise activation. These qualitative observations are consistent with the mechanism-level evidence presented in Section 5.7.

6. Conclusion

This paper proposes BR-DETR, a lightweight real-time detection Transformer for UAV-based concrete bridge defect inspection, designed and evaluated as a measurement system whose localization and dimensional accuracy govern the reliability of downstream condition assessment. The C2f-DIMB backbone replaces fixed-kernel ResNet-18 with dynamic multi-branch depthwise convolutions, enabling input-adaptive receptive-field coverage for heterogeneous defect morphologies. The MSDEP injects fine-grained P_2-level spatial information through an efficient pathway, matching the small-object gain of a genuine P_2 head at approximately one-quarter of the additional FLOPs while preserving embedded real-time throughput. The AIFI-HiLo encoder decomposes self-attention into high-frequency boundary and low-frequency context branches, grounding the frequency decomposition in the spectral characteristics of structural defects.

On a five-class dataset of 8,924 images, BR-DETR improves mAP50 and mAP50-95 over RT-DETR-R18 by 4.67 and 5.67 pp while reducing parameters by 31.9% and GFLOPs by 8.6%, running at 320 FPS; cross-protocol and multi-seed validation confirms the statistical significance

of these margins ($p < 0.001$). Performance gains concentrate on visually challenging categories and small instances, with small-object mAP50 improving by 8.4 pp. From the measurement perspective, BR-DETR reduces the mean center localization error by 28.0% (to 4.92 px, approximately 0.98 mm at an illustrative GSD of 0.2 mm/pixel), lowers dimensional and area quantification errors by 3.2 and 5.1 pp, and raises the share of high-overlap detections (IoU ≥ 0.75) from 52.3% to 61.8%. Deployment on the Jetson Xavier NX at 46 FPS, with 21.7 ± 1.2 ms per-frame latency, a 698 MB memory footprint, 11.2 W average power, and stable 30-minute thermal behavior, demonstrates feasibility for real-time onboard inspection.

Two limitations remain. First, the residual missed-detection rate positions the current system as an early-warning screening instrument rather than a stand-alone replacement for statutory inspection; future work will pursue multi-pass detection fusion and confidence-calibrated re-scanning strategies to close this gap. Second, leave-one-source-out evaluation confirms applicability within the studied bridge population; cross-domain adaptation to bridges with substantially different structural materials, imaging conditions, or geographical settings remains the primary avenue for future extension, to be pursued through domain-adaptive training combined with adverse-condition image-enhancement preprocessing and pixel-level measurement extension, such as crack-width profiling along the defect skeleton.

Declaration of generative AI use.

During the preparation of this manuscript, the authors used AI-assisted writing tools to improve the language clarity and readability of the manuscript. After using these tools, the authors reviewed and edited all content as necessary and take full responsibility for the content of the publication. AI tools were not used for data analysis, figure generation, or the development of the proposed methodology.

CRediT authorship contribution statement

Shuai Zhang: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Qi Chen:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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